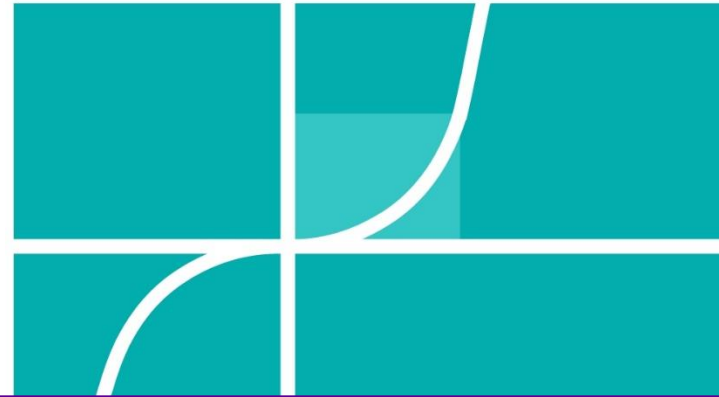


Discrete Structure for Computing:



Sequences and series





- **Sequences**
- The formula for the n^{th} term
- Recurrence relations
- Arithmetic sequences
- Arithmetic series
- The sum of the first n natural numbers
- The sum of an arithmetic series
- Using Σ notation
- Examination-style questions



Arithmetic Sequences and Partial Sums

Arithmetic Sequences

Arithmetic Sequences

- A sequence whose consecutive terms have a common difference is called an **arithmetic sequence**.

Definition of Arithmetic Sequence

A sequence is **arithmetic** when the differences between consecutive terms are the same. So, the sequence

$$a_1, a_2, a_3, a_4, \dots, a_n, \dots$$

is arithmetic when there is a number d such that

$$a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \dots = d.$$

The number d is the **common difference** of the sequence.

Example 1 – *Examples of Arithmetic Sequences*

- **b.** The sequence whose n th term is

- $7 - 5n$

- is arithmetic.

- The common difference between consecutive terms is -5 .

Begin with $n = 1$.

- $2, -3, \overbrace{-8, -13}^{-5}, \dots, 7 - 5n, \dots$

Example 1 – *Examples of Arithmetic Sequences*

- **c.** The sequence whose n th term is

$$\frac{1}{4}(n + 3)$$

- is arithmetic.

$$\frac{1}{4}.$$

- The common difference between consecutive terms is $\frac{1}{4}$.
Begin with $n = 1$.

$$\frac{5}{4} - 1 = \frac{1}{4}$$

Geometric Sequences and Series

A sequence is **geometric** if the ratios of consecutive terms are the same.

2, 8, 32, 128, 512, ...



geometric sequence

$$\frac{8}{2} = 4$$

$$\frac{32}{8} = 4$$

$$\frac{128}{32} = 4$$

$$\frac{512}{128} = 4$$

The **common ratio**, r , is 4.

Example 1.

a. Is the sequence geometric? If so, what is r ?

$$2, 4, 8, 16, \dots 2^n, \dots$$

$$\frac{4}{2} = 2, \quad \frac{8}{4} = 2, \quad \frac{16}{8} = 2 \qquad r = 2$$

b. Is the sequence geometric? If so, what is r ?

$$-\frac{1}{3}, \frac{1}{9}, -\frac{1}{27}, \frac{1}{81}, \dots, \left(-\frac{1}{3}\right)^n, \dots$$

$$\frac{\frac{1}{9}}{-\frac{1}{3}} = -\frac{1}{3} \quad \frac{-\frac{1}{27}}{\frac{1}{9}} = -\frac{1}{3} \quad \frac{\frac{1}{81}}{-\frac{1}{27}} = -\frac{1}{3} \qquad r = -\frac{1}{3}$$

The ***n*th term** of a geometric sequence has the form

$$a_n = a_1 r^{n-1}$$

where r is the common ratio of consecutive terms of the sequence.

The diagram illustrates a geometric sequence with terms 15, 75, 375, 1875, A red bracket above the first two terms (15 and 75) is labeled with the common ratio $r = \frac{75}{15} = 5$. A red arrow points from the first term, 15, to the label $a_1 = 15$. Below the terms, red arrows point from the labels $a_2 = 15(5)$, $a_3 = 15(5^2)$, and $a_4 = 15(5^3)$ to the terms 75, 375, and 1875, respectively.

The n th term is $15(5^{n-1})$.

Example 2. Write the first five terms of the geometric sequence whose first term is $a_1 = 3$ and whose common ratio is $r = 2$.

$$a_1 = 3$$

$$a_2 = 3 \cdot 2^1 = 6$$

$$a_3 = 3 \cdot 2^2 = 12$$

$$a_4 = 3 \cdot 2^3 = 24$$

$$a_5 = 3 \cdot 2^4 = 48$$

Example 3. Find the 15th term of the geometric sequence whose first term is 20 and whose common ratio is 1.05.

$$a_n = a_1 r^{n-1}$$

$$a_{15} = 20(1.05)^{14} \approx 39.599$$

Example 4. Find a formula for the n th term of the following geometric sequence. What is the ninth term of the sequence?

$$5, 15, 45, \dots$$

Find the common ratio

$$15 \div 5 = 3$$

$$45 \div 15 = 3$$

$$a_n = 5(3)^{n-1}$$

$$a_9 = 5(3)^8 = 32,805$$

Example 5. The 4th term of a geometric sequence is 125, and the 10th term is $\frac{125}{64}$.

Find the 14th term. Assume that the terms of the sequence are positive.

$$a_{10} = a_4 r^{10-4}$$

$$\frac{125}{64} = 125r^6$$

$$\frac{1}{64} = r^6$$

$$\frac{1}{2} = r$$

$$a_{14} = a_{10} r^4 = \frac{125}{64} \left(\frac{1}{2} \right)^4 = \frac{125}{1024}$$

The sum of the first n terms of a sequence is represented by **summation notation**.

$$\sum_{i=1}^n a_i = a_1 + a_2 + a_3 + a_4 + \cdots + a_n$$

upper limit of summation (points to n)

lower limit of summation (points to $i=1$)

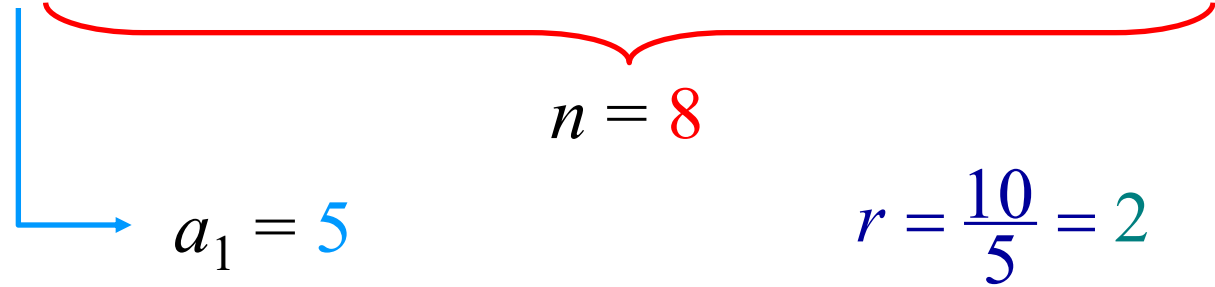
index of summation (points to i)

$$\begin{aligned}\sum_{n=1}^5 4^n &= 4^1 + 4^2 + 4^3 + 4^4 + 4^5 \\ &= 4 + 16 + 64 + 256 + 1024 \\ &= 1364\end{aligned}$$

The **sum of a finite geometric sequence** is given by

$$S_n = \sum_{i=1}^n a_1 r^{i-1} = a_1 \left(\frac{1-r^n}{1-r} \right).$$

$$5 + 10 + 20 + 40 + 80 + 160 + 320 + 640 = ?$$



$n = 8$

$a_1 = 5$

$r = \frac{10}{5} = 2$

$$S_n = a_1 \left(\frac{1-r^n}{1-r} \right) = 5 \left(\frac{1-2^8}{1-2} \right) = 5 \left(\frac{1-256}{1-2} \right) = 5 \left(\frac{-255}{-1} \right) = 1275$$

Example 6. Find the sum $\sum_{n=1}^{12} 4(0.3)^n$

Write out a few terms.

$$\sum_{n=1}^{12} 4(0.3)^n = 4(0.3)^1 + 4(0.3)^2 + 4(0.3)^3 + \dots + 4(0.3)^{12}$$

$$a_1 = 4(0.3) \rightarrow r = 0.3 \text{ and } n = 12$$

$$\begin{aligned} \sum_{n=1}^{12} 4(0.3)^n &= a_1 \left(\frac{1-r^n}{1-r} \right) \\ &= 4(0.3) \left[\frac{1-(0.3)^{12}}{1-0.3} \right] \approx 1.714 \end{aligned}$$

If the index began at $i = 0$, you would have to adjust your formula

$$\sum_{i=0}^{12} 4(0.3)^n = 4(0.3)^0 + \sum_{n=1}^{12} 4(0.3)^n = 4 + \sum_{n=1}^{12} 4(0.3)^n \approx 4 + 1.714 \approx 5.714$$

The sum of the terms of an infinite geometric sequence is called a **geometric series**.

If $|r| < 1$, then the infinite geometric series

$$a_1 + a_1r + a_1r^2 + a_1r^3 + \cdots + a_1r^{n-1} + \cdots$$

has the sum $S = \sum_{i=0}^{\infty} a_1r^i = \frac{a_1}{1-r}$.

If $r \geq |1|$, then the series does not have a sum.

Example 7. Use a graphing calculator to find the first six partial sums of the series. Then find the sum of the series. $\sum_{n=1}^{\infty} 4(0.6)^{n-1}$

$$\text{cum sum}(\text{seq}(4*0.6^{(x-1)}, x, 1, 6))$$

$$4, 6.4, 7.84, 8.704, 9.2224, 9.53344$$

Use the formula for the sum of an infinite series to find the sum.

$$S = \frac{4}{1 - 0.6} = 10$$

Example 8. Find the sum of $3 + 0.3 + 0.03 + 0.003 + \dots$,

$$S = \frac{3}{1 - 0.1} = 3.33$$

Arithmetic Sequences

The n th Term of an Arithmetic Sequence

The n th term of an arithmetic sequence has the form

$$a_n = a_1 + (n - 1)d$$

where d is the common difference between consecutive terms of the sequence and a_1 is the first term of the sequence.

Sequence

- A sequence is a list of objects arranged in a definite order
- Difference between set and sequence
 - A set has no order and no duplicated elements
 - A sequence has a specific order and elements may be duplicated
- Nomenclature: $a_1, a_2, \dots, a_n$

Sequence Examples

- 1, 2, 3, 2, 2, 3, 1 is a sequence, but not a set
- The sequences 1, 2, 3, 2, 2, 3, 1 and 2, 2, 1, 3 are made from elements of the set $\{1, 2, 3\}$
- The sequences 1, 2, 3, 2, 2, 3, 1 and 2, 1, 3, 2, 2, 3, 1 only switch two elements, but that alone makes them unequal

Types of Sequences

- Sequence may stop after n steps (***finite***) or go on for ever (***infinite***)
- Formulas can be used to describe sequences
 - Recursive
 - Explicit

Recursive Sequences

- In a recursive sequence, the next item in the sequence is determined from previous values
- Difficult to determine say 100th element since previous 99 need to be determined first.

- Example:

$$a_1 = 1$$

$$a_2 = 2$$

$$a_n = a_{n-1} + a_{n-2}$$

Explicit Sequences

- In an explicit sequence, the n th item is determined by a formula depending only on n
- Easier to determine any element
- Example: $a_n = 2 \times n$

Linear Array

- Principles of sequences can be applied to computers, specifically, arrays. There are some differences though.
- Sequence
 - Well-defined
 - Modification of any element or its order creates new sequence
- Array
 - May not have all elements initialized
 - Modification of array by software may occur
 - Even if the array has variable length, we're ultimately limited to finite length

Countable and Uncountable

- A **set** is **countable** if it corresponds to some sequence.
 - Members of set can be arranged in a list
 - Elements have position
- All finite sets are countable
- Not all infinite sets are countable, and are therefore **uncountable**
 - Best example is real numbers
 - E.g., what comes after 1.23534?

Concatenation

- Two strings may be joined into a single string
- Assume $w_1 = s_1s_2s_3s_4...s_n$ and $w_2 = t_1t_2t_3t_4...t_k$
- The catenation of w_1 and w_2 is the sequence $s_1s_2s_3s_4...s_nt_1t_2t_3t_4...t_k$
- Notation: Concatenation of w_1 and w_2 is written as $w_1 \cdot w_2$ or w_1w_2

Common Sequences

Arithmetic	$a, a+d, a+2d, a+3d, a+4d, \dots$
n^2	$1, 4, 9, 16, 25, \dots$
n^3	$1, 8, 27, 64, 125, \dots$
n^4	$1, 16, 81, 256, 625, \dots$
2^n	$2, 4, 8, 16, 32, \dots$
3^n	$3, 9, 27, 81, 243, \dots$
$n!$	$1, 2, 6, 24, 120, \dots$

Summations

- Notation for describing the sum of the terms $a_m, a_{m+1}, \dots, a_n$ from the sequence, $\{a_n\}$

$$a_m + a_{m+1} + \dots + a_n = \sum_{j=m}^n a_j$$

- j is the index of summation (dummy variable)
- The index of summation runs through all integers from its lower limit, m , to its upper limit, n .

Examples

$$\sum_{j=1}^5 j = \sum_{j=0}^4 (j+1) = 1+2+3+4+5 = 15$$

$$\sum_{j=1}^5 1/j = 1 + 1/2 + 1/3 + 1/4 + 1/5$$

$$1 + \sum_{j=2}^5 1/j = 1 + 1/2 + 1/3 + 1/4 + 1/5$$

Summations follow all the rules of multiplication and addition!

$$c \sum_{j=1}^n j = \sum_{j=1}^n cj = c(1+2+\dots+n) = c + 2c + \dots + nc$$

$$r \sum_{j=0}^n ar^j = \sum_{j=0}^n ar^{j+1} = \sum_{k=1}^{n+1} ar^k =$$

$$ar^{n+1} + \sum_{k=1}^n ar^k = ar^{n+1} - a + \sum_{k=0}^n ar^k$$

Telescoping Sums

$$\sum_{j=1}^n (a_j - a_{j-1}) = (a_1 - a_0) + (a_2 - a_1) +$$

$$(a_3 - a_2) + \dots + (a_n - a_{n-1}) = a_n - a_0$$

Example

$$\sum_{k=1}^4 [k^2 - (k-1)^2] =$$

$$(1^2 - 0^2) + (2^2 - 1^2) + (3^2 - 2^2) + (4^2 - 3^2)$$

$$4^2 = 16 - 0 = 16$$

Closed Form Solutions

A simple formula that can be used to calculate a sum without doing all the additions.

Example:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Proof: First we note that $k^2 - (k-1)^2 = k^2 - (k^2 - 2k + 1) = 2k - 1$.
Since $k^2 - (k-1)^2 = 2k - 1$, then we can sum each side from $k=1$ to $k=n$

$$\sum_{k=1}^n [k^2 - (k-1)^2] = \sum_{k=1}^n (2k - 1)$$

Proof (cont.)

$$\sum_{k=1}^n [k^2 - (k-1)^2] = \sum_{k=1}^n (2k-1)$$

$$\sum_{k=1}^n [k^2 - (k-1)^2] = \sum_{k=1}^n 2k + \sum_{k=1}^n (-1)$$

$$n^2 - 0^2 = 2 \sum_{k=1}^n (k) + -n$$

$$n^2 + n = 2 \sum_{k=1}^n (k)$$

$$\sum_{k=1}^n k = \frac{n^2 + n}{2}$$

Closed Form Solutions to Sums

$$\sum_{j=0}^n j = 0 + 1 + \dots + n = n(n+1)/2$$

$$\sum_{j=0}^n j^2 = 0^2 + 1^2 + \dots + n^2 = n(n+1)(2n+1)/6$$

$$\sum_{k=1}^n k^3 = \frac{n^2(n+1)^2}{4}$$

$$\sum_{k=0}^n ar^k = \frac{ar^{n+1} - a}{r - 1}, r \neq 1$$

Double Summations

$$\sum_{i=1}^4 \sum_{j=1}^3 ij = \sum_{i=1}^4 \left[\sum_{j=1}^3 ij \right] = \sum_{i=1}^4 (i + 2i + 3i) = \sum_{i=1}^4 6i =$$
$$6 + 12 + 18 + 24 = 60$$

Induction and recursion

Recursion

Mathematical Induction

Section 5.1

Section Summary

- Mathematical Induction
- Examples of Proof by Mathematical Induction
- Mistaken Proofs by Mathematical Induction
- Guidelines for Proofs by Mathematical Induction

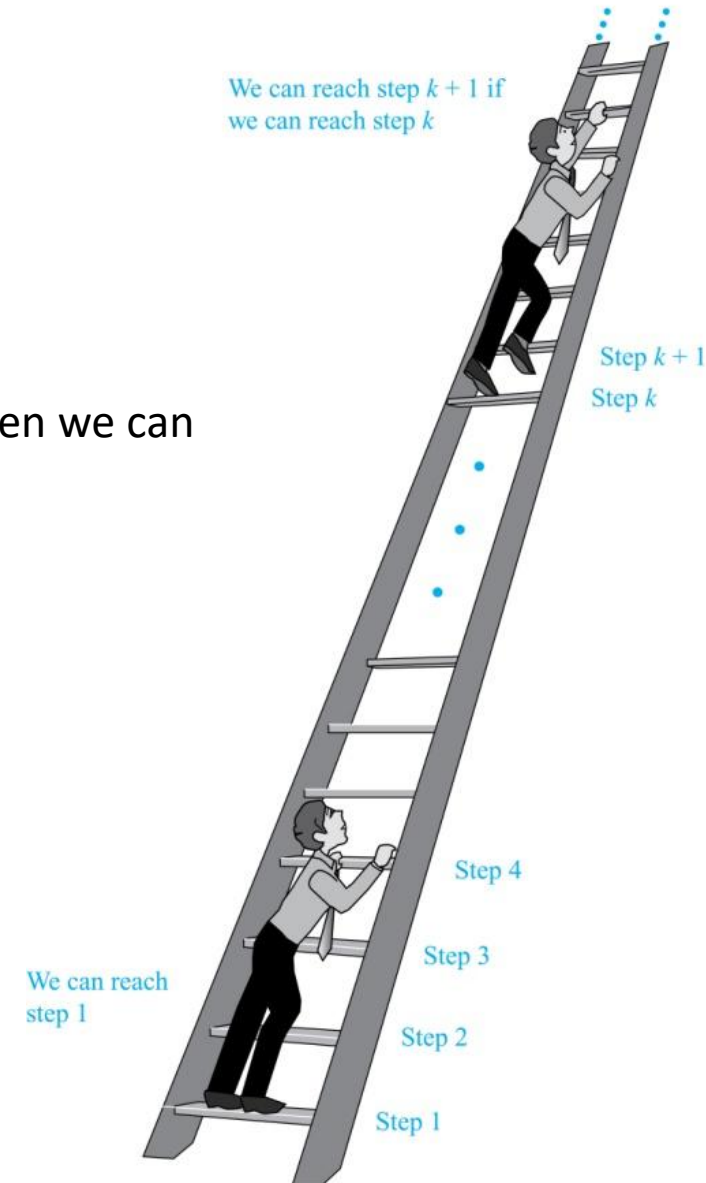
Climbing an Infinite Ladder

Suppose we have an infinite ladder:

1. We can reach the first rung of the ladder.
2. If we can reach a particular rung of the ladder, then we can reach the next rung.

From (1), we can reach the first rung. Then by applying (2), we can reach the second rung. Applying (2) again, the third rung. And so on. We can apply (2) any number of times to reach any particular rung, no matter how high up.

This example motivates proof by mathematical induction.



Principle of Mathematical Induction

Principle of Mathematical Induction: To prove that $P(n)$ is true for all positive integers n , we complete these steps:

- *Basis Step:* Show that $P(1)$ is true.
- *Inductive Step:* Show that $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

To complete the inductive step, assuming the *inductive hypothesis* that $P(k)$ holds for an arbitrary integer k , show that $P(k + 1)$ must be true.

Climbing an Infinite Ladder Example:

- BASIS STEP: By (1), we can reach rung 1.
- INDUCTIVE STEP: Assume the inductive hypothesis that we can reach rung k . Then by (2), we can reach rung $k + 1$.

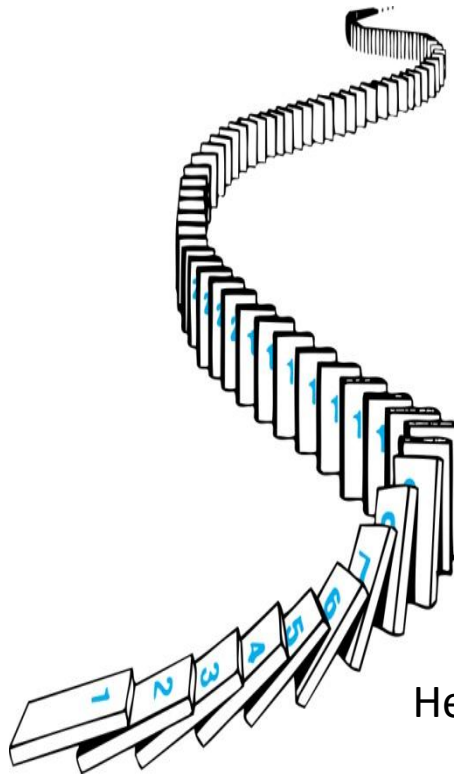
Hence, $P(k) \rightarrow P(k + 1)$ is true for all positive integers k . We can reach every rung on the ladder.



Remembering How Mathematical Induction Works

Consider an infinite sequence of dominoes, labeled $1, 2, 3, \dots$, where each domino is standing.

Let $P(n)$ be the proposition that the n th domino is knocked over.



We know that the first domino is knocked down, i.e., $P(1)$ is true.

We also know that if whenever the k th domino is knocked over, it knocks over the $(k + 1)$ st domino, i.e, $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

Hence, all dominos are knocked over.

$P(n)$ is true for all positive integers n .

Proving a Summation Formula

Example: Show that: $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

Solution:

- BASIS STEP: $P(1)$ is true since $1(1+1)/2 = 1$.
- INDUCTIVE STEP: Assume the inductive hypothesis holds $P(k)$ and then show that $P(k+1)$ holds as well.

The inductive hypothesis is $\sum_{i=1}^k i = \frac{k(k+1)}{2}$
Under this assumption,

$$\begin{aligned} 1 + 2 + \dots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$



Conjecturing and Proving Correct a Summation Formula

Example: Conjecture and prove correct a formula for the sum of the first n positive odd integers. Then prove your conjecture.

Solution: We have: $1 = 1$, $1 + 3 = 4$, $1 + 3 + 5 = 9$, $1 + 3 + 5 + 7 = 16$, $1 + 3 + 5 + 7 + 9 = 25$.

- We can conjecture that the sum of the first n positive odd integers is n^2 ,

- We prove the conjecture is proved correct with mathematical induction.
- BASIS STEP: $P(1)$ is true since $1^2 = 1$.
- INDUCTIVE STEP: $P(k) \rightarrow P(k + 1)$ for every positive integer k .

Assume the inductive hypothesis holds and then show that $P(k + 1)$ holds as well.

Inductive Hypothesis: $1 + 3 + 5 + \cdots + (2k - 1) = k^2$

- So, assuming $P(k)$, it follows that:

$$\begin{aligned} 1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1) &= [1 + 3 + 5 + \cdots + (2k - 1)] + (2k + 1) \\ &= k^2 + (2k + 1) \text{ (by the inductive hypothesis)} \\ &= k^2 + 2k + 1 \\ &= (k + 1)^2 \end{aligned}$$

- Hence, we have shown that $P(k + 1)$ follows from $P(k)$. Therefore the sum of the first n positive odd integers is n^2 .



Important Points About Using Mathematical Induction

- Mathematical induction can be expressed as the rule of inference

$(P(1) \wedge \forall k (P(k) \rightarrow P(k + 1))) \rightarrow \forall n P(n),$
where the domain is the set of positive integers.

- In a proof by mathematical induction, we don't assume that $P(k)$ is true for all positive integers! We show that if we assume that $P(k)$ is true, then $P(k + 1)$ must also be true.
- Proofs by mathematical induction do not always start at the integer 1. In such a case, the basis step begins at a starting point b where b is an integer. We will see examples of this soon.

Validity of Mathematical Induction

$$(P(1) \wedge \forall k (P(k) \rightarrow P(k + 1))) \rightarrow \forall n P(n)$$

- Mathematical induction is valid because of the well ordering property, which states that every nonempty subset of the set of positive integers has a least element (*see Section 5.2 and Appendix 1*). Here is the proof:
 - Suppose that $P(1)$ holds and $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .
 - Assume there is at least one positive integer n for which $P(n)$ is false. Then the set S of positive integers for which $P(n)$ is false is nonempty.
 - By the well-ordering property, S has a least element, say m .
 - We know that m can not be 1 since $P(1)$ holds.
 - Since m is positive and greater than 1, $m - 1$ must be a positive integer. Since $m - 1 < m$, it is not in S , so $P(m - 1)$ must be true.
 - But then, since the conditional $P(k) \rightarrow P(k + 1)$ for every positive integer k holds, $P(m)$ must also be true. This contradicts $P(m)$ being false.
 - Hence, $P(n)$ must be true for every positive integer n .

Proving Inequalities

Example: Use mathematical induction to prove that $n < 2^n$ for all positive integers n .

Solution: Let $P(n)$ be the proposition that $n < 2^n$.

- BASIS STEP: $P(1)$ is true since $1 < 2^1 = 2$.
- INDUCTIVE STEP: Assume $P(k)$ holds, i.e., $k < 2^k$, for an arbitrary positive integer k .
- Must show that $P(k + 1)$ holds. Since by the inductive hypothesis, $k < 2^k$, it follows that:

$$k + 1 < 2^k + 1 \leq 2^k + 2^k = 2^k (1+1) = 2 \cdot 2^k = 2^{k+1}$$

Therefore $n < 2^n$ holds for all positive integers n .



Proving Inequalities

Example: Use mathematical induction to prove that $2^n < n!$, for every integer $n \geq 4$.

Solution: Let $P(n)$ be the proposition that $2^n < n!$.

- BASIS STEP: $P(4)$ is true since $2^4 = 16 < 4! = 24$.
- INDUCTIVE STEP: Assume $P(k)$ holds, i.e., $2^k < k!$ for an arbitrary integer $k \geq 4$. To show that $P(k + 1)$ holds:

$$\begin{aligned} 2^{k+1} &= 2 \cdot 2^k \\ &< 2 \cdot k! && \text{(by the inductive hypothesis)} \\ &< (k + 1)k! \\ &= (k + 1)! \end{aligned}$$

Therefore, $2^n < n!$ holds, for every integer $n \geq 4$. 

Note that here the basis step is $P(4)$, since $P(0)$, $P(1)$, $P(2)$, and $P(3)$ are all false.

Proving Divisibility Results

Example: Use mathematical induction to prove that $n^3 - n$ is divisible by 3, for every positive integer n .

Solution: Let $P(n)$ be the proposition that $n^3 - n$ is divisible by 3.

- BASIS STEP: $P(1)$ is true since $1^3 - 1 = 0$, which is divisible by 3.
- INDUCTIVE STEP: Assume $P(k)$ holds, i.e., $k^3 - k$ is divisible by 3, for an arbitrary positive integer k . To show that $P(k + 1)$ follows:

$$\begin{aligned}(k + 1)^3 - (k + 1) &= (k^3 + 3k^2 + 3k + 1) - (k + 1) \\ &= (k^3 - k) + 3(k^2 + k)\end{aligned}$$

By the inductive hypothesis, the first term $(k^3 - k)$ is divisible by 3 and the second term is divisible by 3 since it is an integer multiplied by 3. So by part (i) of Theorem 1 in Section 4.1, $(k + 1)^3 - (k + 1)$ is divisible by 3.

Therefore, $n^3 - n$ is divisible by 3, for every positive integer n .



Guidelines: Mathematical Induction Proofs

Template for Proofs by Mathematical Induction

1. Express the statement that is to be proved in the form “for all $n \geq b$, $P(n)$ ” for a fixed integer b .
2. Write out the words “Basis Step.” Then show that $P(b)$ is true, taking care that the correct value of b is used. This completes the first part of the proof.
3. Write out the words “Inductive Step.”
4. State, and clearly identify, the inductive hypothesis, in the form “assume that $P(k)$ is true for an arbitrary fixed integer $k \geq b$.”
5. State what needs to be proved under the assumption that the inductive hypothesis is true. That is, write out what $P(k + 1)$ says.
6. Prove the statement $P(k + 1)$ making use the assumption $P(k)$. Be sure that your proof is valid for all integers k with $k \geq b$, taking care that the proof works for small values of k , including $k = b$.
7. Clearly identify the conclusion of the inductive step, such as by saying “this completes the inductive step.”
8. After completing the basis step and the inductive step, state the conclusion, namely that by mathematical induction, $P(n)$ is true for all integers n with $n \geq b$.

Introduction

- A geometric sequence is a list of terms separated by a **common ratio**, r , which is the number multiplied by each consecutive term in a geometric sequence.
- A geometric sequence is an **exponential function** with a domain of **whole numbers** in which the ratio between any two consecutive terms is equal.



Introduction (*continued*)

Just like arithmetic sequences, **Geometric sequences** can be represented by formulas, either explicit or recursive, and those formulas can be used to find a certain term of the sequence or the number of a certain value in the sequence.

Recall

- A **recursive formula** is a formula used to find the next term of a sequence when the previous term is known.
- An **explicit formula** is a formula used to find the n th term of a sequence.



Formulas and their Purpose

Geometric Sequences

Explicit Formula: $A_n = a_1 \cdot r^{n-1}$

“Finds a specific term”

First Term

Current
Term

Common Ratio

Recursive Formula: $A_n = A_{n-1} \cdot r$

Previous
Term

“Uses previous terms to find the next terms”

Steps to create formulas and solve for geometric sequences

1. Find the common ratio by dividing the 2nd term by the 1st term.
2. Decide which formula to use. (explicit or recursive)
3. Substitute your values to create your formula.
4. Find the specific term if asked to do so.



Guided Practice

Example 1

Create the recursive formula that defines the sequence:

A geometric sequence is defined by
 $2, 8, 32, 128, \dots$



Guided Practice

Example 1, *continued*

Create the recursive formula that defines the sequence:

A geometric sequence is defined by 2, 8, 32, 128, ...

Step 1: Find the common ratio.

$$\frac{\text{2nd term}}{\text{1st term}} = \frac{A_2}{A_1} = \frac{8}{2} = 4$$

Step 3: Substitute what you have.

Since $r = 4$ then

$$A_n = A_{n-1} \cdot 4$$

Step 2: Explicit or Recursive Formula?

We will use the recursive formula which is

$$A_n = A_{n-1} \cdot r$$



Guided Practice

Example 2

Create the recursive formula that defines the sequence:

A geometric sequence is defined by

$$45, -15, 5, -\frac{5}{3}, \dots$$



Guided Practice

Example 2, *continued*

Create the recursive formula that defines the sequence:

A geometric sequence is defined by 45, -15, 5, $-\frac{5}{3}$, ...

Step 1: Find the common ratio	Step 3: Substitute what you have
Step 2: Explicit or Recursive Formula?	



You Try 1

Use the following sequence to create a recursive formula. $A_n = A_{n-1} \cdot r$

10, -30, 90, -270, ...

Step 1: Find the common ratio	Step 3: Substitute what you have
Step 2: Explicit or Recursive Formula?	



Guided Practice

Example 3

A geometric sequence is defined recursively by $a_n = a_{n-1} \cdot -3$, with $a_1 = 6$. Find the first 5 terms of the sequence.

Using the recursive formula:

$$a_1 = 6$$

$$a_2 = a_1 \cdot -3$$

$$a_2 =$$

$$a_3 =$$

$$a_4 =$$

$$a_5 =$$

The first five terms of the sequence are:



Guided Practice – Example 4

A geometric sequence is defined recursively by $a_n = a_{n-1} \cdot \frac{1}{10}$, with $a_1 = 3000$. Find the first 5 terms of the sequence.

Using the recursive formula:

$$a_1 = 3000$$

$$a_2 = a_1 \cdot \frac{1}{10}$$

$$a_2 = 3000 \cdot \frac{1}{10} = 300$$

$$a_3 = 300 \cdot \frac{1}{10} = 30$$

$$a_4 = 30 \cdot \frac{1}{10} = 3$$

$$a_5 = 3 \cdot \frac{1}{10} = \frac{3}{10}$$

The first five terms of the sequence are:

3000, 300, 30, 3, and $\frac{3}{10}$.



You Try 2

An arithmetic sequence is defined recursively by

$$a_n = a_{n-1} \cdot 6, \text{ with } a_1 = 0.2$$

Find the first 5 terms of the sequence.



Guided Practice

Example 5

Write an explicit formula to represent the sequence from example 1, and find the 10th term.

The first five terms of the sequence are:
2, 8, 32, 128, and 512.



Guided Practice: **Example 5, continued**

The first five terms of the sequence are:
2, 8, 32, 128, and 512.

Step 1: Find the common ratio & a_1. $\frac{\text{2nd term}}{\text{1st term}} = \frac{a_2}{a_1} = \frac{8}{2} = 4$ and a_1 is the first term which is 2.	Step 3: Substitute what you have. $A_n = 2 \cdot 4^{n-1}$
Step 2: Explicit or Recursive Formula? We will use the explicit formula since we are finding a specific term. $A_n = a_1 \cdot r^{n-1}$	Step 4: Evaluate for specific term. $\begin{aligned} A_{10} &= 2 \cdot 4^{10-1} \\ &= 2 \cdot 4^9 \\ &= 524,288 \end{aligned}$ So 524,288 is the 10th term in the sequence.



Guided Practice

Example 6

Write an explicit formula to represent the sequence from example 3, and find the 15th term.

The first five terms of the sequence are:

6, -18, 54, -162, and 486



Recursively Defined Functions

Example: Suppose f is defined by:

$$f(0) = 3,$$

$$f(n + 1) = 2f(n) + 3$$

Find $f(1), f(2), f(3), f(4)$

Solution:

- $f(1) = 2f(0) + 3 = 2 \cdot 3 + 3 = 9$
- $f(2) = 2f(1) + 3 = 2 \cdot 9 + 3 = 21$
- $f(3) = 2f(2) + 3 = 2 \cdot 21 + 3 = 45$
- $f(4) = 2f(3) + 3 = 2 \cdot 45 + 3 = 93$

Example: Give a recursive definition of the factorial function $n!$:

Solution:

$$f(0) = 1$$

$$f(n + 1) = (n + 1) \cdot f(n)$$

Recursively Defined Functions

Example: Give a recursive definition of:

$$\sum_{k=0}^n a_k.$$

Solution: The first part of the definition is

$$\sum_{k=0}^0 a_k = a_0.$$

The second part is

$$\sum_{k=0}^{n+1} a_k = \left(\sum_{k=0}^n a_k \right) + a_{n+1}.$$

Fibonacci Numbers

Fibonacci
(1170- 1250)



Example : The Fibonacci numbers are defined as follows:

$$f_0 = 0$$

$$f_1 = 1$$

$$f_n = f_{n-1} + f_{n-2}$$

Find f_2, f_3, f_4, f_5 .

- $f_2 = f_1 + f_0 = 1 + 0 = 1$
- $f_3 = f_2 + f_1 = 1 + 1 = 2$
- $f_4 = f_3 + f_2 = 2 + 1 = 3$
- $f_5 = f_4 + f_3 = 3 + 2 = 5$

In Chapter 8, the Fibonacci numbers are used to model population growth of rabbits. This was an application described by Fibonacci himself.

Recursively Defined Sets and Structures

Recursive definitions of sets have two parts:

- The *basis step* specifies an initial collection of elements.
- The *recursive step* gives the rules for forming new elements in the set from those already known to be in the set.
- Sometimes the recursive definition has an *exclusion rule*, which specifies that the set contains nothing other than those elements specified in the basis step and generated by applications of the rules in the recursive step.
- We will always assume that the exclusion rule holds, even if it is not explicitly mentioned.
- We will later develop a form of induction, called *structural induction*, to prove results about recursively defined sets.

Recursively Defined Sets and Structures

Example : Subset of Integers S :

BASIS STEP: $3 \in S$.

RECURSIVE STEP: If $x \in S$ and $y \in S$, then $x + y$ is in S .

- Initially 3 is in S , then $3 + 3 = 6$, then $3 + 6 = 9$, etc.

Example: The natural numbers $\mathbf{N}$.

BASIS STEP: $0 \in \mathbf{N}$.

RECURSIVE STEP: If n is in $\mathbf{N}$, then $n + 1$ is in $\mathbf{N}$.

- Initially 0 is in S , then $0 + 1 = 1$, then $1 + 1 = 2$, etc.

String Concatenation

Definition: Two strings can be combined via the operation of *concatenation*. Let Σ be a set of symbols and Σ^* be the set of strings formed from the symbols in Σ . We can define the concatenation of two strings, denoted by $\cdot$, recursively as follows.

BASIS STEP: If $w \in \Sigma^*$, then $w \cdot \lambda = w$.

RECURSIVE STEP: If $w_1 \in \Sigma^*$ and $w_2 \in \Sigma^*$ and $x \in \Sigma$, then $w_1 \cdot (w_2 x) = (w_1 \cdot w_2)x$.

- Often $w_1 \cdot w_2$ is written as $w_1 w_2$.
- If $w_1 = abra$ and $w_2 = cadabra$, the concatenation $w_1 w_2 = abracadabra$.

Length of a String

Example: Give a recursive definition of $l(w)$, the length of the string w .

Solution: The length of a string can be recursively defined by:

$$l(\lambda) = 0;$$

$$l(wx) = l(w) + 1 \text{ if } w \in \Sigma^* \text{ and } x \in \Sigma.$$

Recursive Factorial Algorithm

Example: Give a recursive algorithm for computing $n!$, where n is a nonnegative integer.

- **Solution:** Use the recursive definition of the factorial function: $n! = n \cdot (n - 1)!$.

```
procedure factorial( $n$ : nonnegative integer)
  if  $n = 0$  then return 1
  else return  $n * \text{factorial}(n - 1)$ 
  {output is  $n!$ }
```

E.g.: $4! = 4 * 3! = 4 * 3 * 2! = 4 * 3 * 2 * 1! = 4 * 3 * 2 * 1 * 0! = 4 * 3 * 2 * 1 * 1 = 24$

Recursive Exponentiation Algorithm

Example: Give a recursive algorithm for computing a^n , where a is a nonzero real number and n is a nonnegative integer.

Solution: Use the recursive definition of a^n .

```
procedure power( $a$ : nonzero real number,  $n$ : nonnegative integer)
  if  $n = 0$  then return 1
  else return  $a * \text{power}(a, n - 1)$ 
  {output is  $a^n$ }
```

Recursive GCD Algorithm

Example: Give a recursive algorithm for computing the greatest common divisor of two nonnegative integers a and b with $a < b$.

Solution: Use the reduction

$$\gcd(a, b) = \gcd(b \bmod a, a)$$

and the condition $\gcd(0, b) = b$ when $b > 0$.

```
procedure gcd( $a, b$ : nonnegative integers  
             with  $a < b$ )  
if  $a = 0$  then return  $b$   
else return gcd( $b \bmod a, a$ )  
{output is  $\gcd(a, b)$ }
```

E.g.: $\gcd(5, 8) =$

Proving Recursive Algorithms Correct

- Both mathematical and strong induction are useful techniques to show that recursive algorithms always produce the correct output.

Example: Prove that the algorithm for computing the powers of real numbers is correct.

```
procedure power(a: nonzero real number, n: nonnegative integer)
if n = 0 then return 1
else return a · power(a, n − 1)
{output is  $a^n$ }
```

Solution: Use mathematical induction on the exponent n .

BASIS STEP: $a^0 = 1$ for every nonzero real number a , and $\text{power}(a, 0) = 1$.

INDUCTIVE STEP: The inductive hypothesis is that $\text{power}(a, k) = a^k$, for all $a \neq 0$.
Assuming the inductive hypothesis, the algorithm correctly computes a^{k+1} , since

$$\text{power}(a, k + 1) = a \cdot \text{power}(a, k) = a \cdot a^k = a^{k+1}.$$



Recursion and Iteration

- A **recursive definition** expresses the value of a function at a positive integer in terms of the values of the function at smaller integers.
- Instead of successively reducing the computation to the evaluation of the function at smaller integers, we can start with the value of the function at one or more integers, the base cases, and successively apply the recursive definition to find the values of the function at successive larger integers. Such a procedure is called **iterative**.
- Often an iterative approach for the evaluation of a recursively defined sequence requires much less computation than a procedure using recursion. This is illustrated by the iterative and recursive procedures for finding the *n*th *Fibonacci* number.

Example 5 – *Finding the Sum of a Finite Arithmetic Sequence*

- Find the sum:

$$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19.$$

- **Solution:**

- To begin, notice that the sequence is arithmetic (with a common difference of 2). Moreover, the sequence has 10 terms. So, the sum of the sequence is

- $S_n = 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19$

- $$= \frac{n}{2}(a_1 + a_n)$$
 Formula for sum of an arithmetic sequence

Example 5 – *Solution*

$$= \frac{10}{2}(1 + 19)$$

Substitute 10 for n , 1 for a_1 , and 19 for a_n .

Simplify.

- $= 5(20)$
- $= 100.$

The Sum of a Finite Arithmetic Sequence

- The sum of the first n terms of an infinite sequence is called the **n th partial sum**.
- The n th partial sum of an arithmetic sequence can be found by using the formula for the sum of a finite arithmetic sequence which is on slide #10.

Recursive Definitions

Recursion is a principle closely related to mathematical induction.

In a **recursive definition**, an object is defined in terms of itself.

We can recursively define **sequences, functions and sets**.

Recursively Defined Sequences

Example:

The sequence $\{a_n\}$ of powers of 2 is given by $a_n = 2^n$ for $n = 0, 1, 2, \dots$.

The same sequence can also be defined **recursively**:

$$a_0 = 1$$

$$a_{n+1} = 2a_n \quad \text{for } n = 0, 1, 2, \dots$$

Obviously, induction and recursion are similar principles.

Recursively Defined Functions

We can use the following method to define a function with the **natural numbers** as its domain:

Base case: Specify the value of the function at zero.

Recursion: Give a rule for finding its value at any integer from its values at smaller integers.

Such a definition is called **recursive** or **inductive definition**.

Recursively Defined Functions

Example:

$$f(0) = 3$$

$$f(n + 1) = 2f(n) + 3$$

$$f(0) = 3$$

$$f(1) = 2f(0) + 3 = 2 \cdot 3 + 3 = 9$$

$$f(2) = 2f(1) + 3 = 2 \cdot 9 + 3 = 21$$

$$f(3) = 2f(2) + 3 = 2 \cdot 21 + 3 = 45$$

$$f(4) = 2f(3) + 3 = 2 \cdot 45 + 3 = 93$$

Recursively Defined Functions

How can we recursively define the factorial function
 $f(n) = n!$?

$$f(0) = 1$$

$$f(n + 1) = (n + 1)f(n)$$

$$f(0) = 1$$

$$f(1) = 1f(0) = 1 \cdot 1 = 1$$

$$f(2) = 2f(1) = 2 \cdot 1 = 2$$

$$f(3) = 3f(2) = 3 \cdot 2 = 6$$

$$f(4) = 4f(3) = 4 \cdot 6 = 24$$

Recursively Defined Functions

A famous example: The Fibonacci numbers

$$f(0) = 0, f(1) = 1$$

$$f(n) = f(n - 1) + f(n - 2)$$

$$f(0) = 0$$

$$f(1) = 1$$

$$f(2) = f(1) + f(0) = 1 + 0 = 1$$

$$f(3) = f(2) + f(1) = 1 + 1 = 2$$

$$f(4) = f(3) + f(2) = 2 + 1 = 3$$

$$f(5) = f(4) + f(3) = 3 + 2 = 5$$

$$f(6) = f(5) + f(4) = 5 + 3 = 8$$

Recursively Defined Sets

If we want to recursively define a set, we need to provide two things:

an **initial set** of elements,

rules for the construction of **additional** elements from elements in the set.

Example: Let S be recursively defined by:

$$3 \in S$$

$$(x + y) \in S \text{ if } (x \in S) \text{ and } (y \in S)$$

S is the set of positive integers divisible by 3.

Recursively Defined Sets

Proof:

Let A be the set of all positive integers divisible by 3.

To show that $A = S$, we must show that
 $A \subseteq S$ and $S \subseteq A$.

Part I: To prove that $A \subseteq S$, we must show that every positive integer divisible by 3 is in S .

We will use mathematical induction to show this.

Recursively Defined Sets

Let $P(n)$ be the statement “ $3n$ belongs to S ”.

Basis step: $P(1)$ is true, because 3 is in S .

Inductive step: To show:
If $P(n)$ is true, then $P(n + 1)$ is true.

Assume $3n$ is in S . Since $3n$ is in S and 3 is in S , it follows from the recursive definition of S that $3n + 3 = 3(n + 1)$ is also in S .

Conclusion of Part I: $A \subseteq S$.

Recursively Defined Sets

Part II: To show: $S \subseteq A$.

Basis step: To show:

All initial elements of S are in A . 3 is in A . True.

Inductive step: To show:

If x and y in S are in A , then $(x + y)$ is in A .

Since x and y are both in A , it follows that $3 \mid x$ and $3 \mid y$.

From Theorem 1, Section 2.3, it follows that $3 \mid (x + y)$.

Conclusion of Part II: $S \subseteq A$.

Overall conclusion: $A = S$.

Recursively Defined Sets

Another example:

The well-formed formulae of variables, numerals and operators from $\{+, -, *, /, ^\}$ are defined by:

x is a well-formed formula if x is a numeral or variable.

$(f + g)$, $(f - g)$, $(f * g)$, (f / g) , $(f ^ g)$ are well-formed formulae if f and g are.

Recursive Algorithms

An algorithm is called **recursive** if it solves a problem by reducing it to an instance of the same problem with smaller input.

Example 1: Recursive Euclidean Algorithm

```
procedure gcd(a, b: nonnegative integers with  $a < b$ )  
if  $a = 0$  then gcd(a, b) := b  
else gcd(a, b) := gcd(b mod a, a)
```

Recursive Algorithms

Example II: Recursive Fibonacci Algorithm

procedure fibo(n : nonnegative integer)

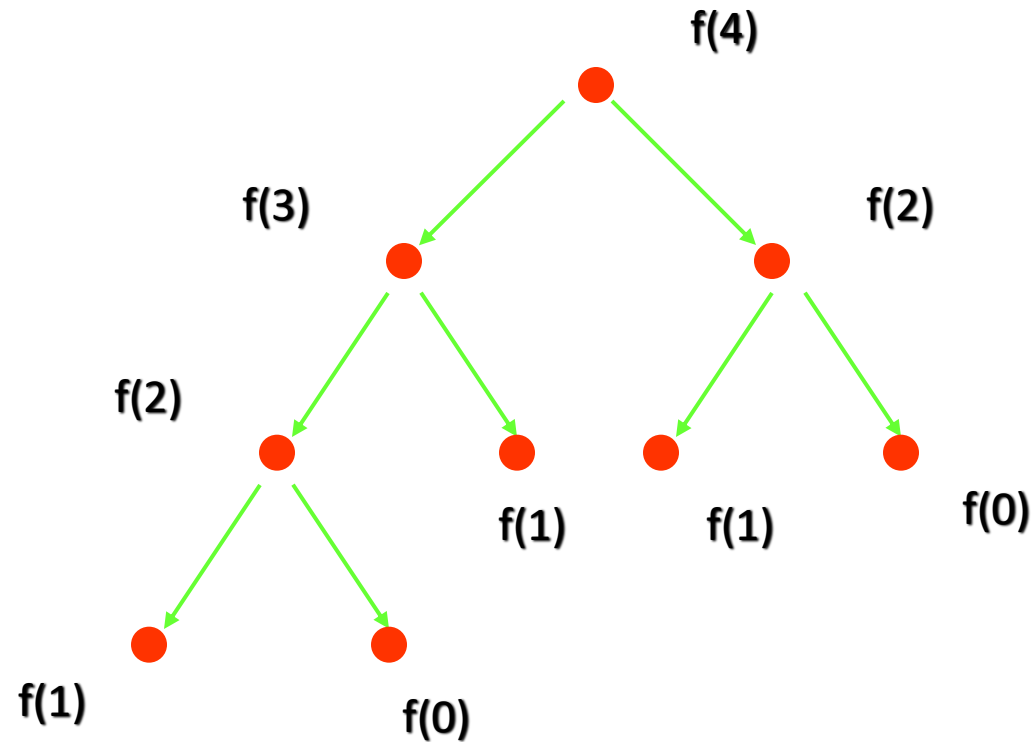
if $n = 0$ **then** fibo(0) := 0

else if $n = 1$ **then** fibo(1) := 1

else fibo(n) := fibo($n - 1$) + fibo($n - 2$)

Recursive Algorithms

Recursive Fibonacci Evaluation:



Recurrence relations

This sequence can also be defined by a **recurrence relation**.

To define a sequence using a recurrence relation we need the value of the first term and an expression relating each term to a previous term.

For the sequence $-1, 3, 7, 11, 15, \dots$, each term can be found by adding 4 to the previous term.

We can write:

$$u_1 = -1$$

$$u_2 = u_1 + 4 = 3$$

$$u_3 = u_2 + 4 = 7$$

$$u_4 = u_3 + 4 = 11 \quad \text{and so on.}$$

In general:

$$u_{n+1} = u_n + 4$$



Recurrence relations

A recurrence relation together with the first term of a sequence is called an **inductive definition**.

So the inductive definition for the sequence $-1, 3, 7, 11, 15, \dots$ is $u_1 = -1, u_{n+1} = u_n + 4$.

A sequence is given by the recurrence relation $u_{n+1} = 2u_n + 1$ with $u_1 = 3$. Write down the first five terms of the sequence.

$$u_1 = 3$$

$$u_2 = (2 \times 3) + 1 = 7$$

$$u_3 = (2 \times 7) + 1 = 15$$

$$u_4 = (2 \times 15) + 1 = 31$$

$$u_5 = (2 \times 31) + 1 = 63$$

So the first five terms in the sequence are 3, 7, 15, 31 and 63.



Using an inductive definition



Arithmetic sequences

In an **arithmetic sequence** (or **arithmetic progression**) the difference between any two consecutive terms is always the same. This is called the **common difference**.

For example, the sequence:

$$8, 11, 14, 17, 20, \dots$$

is an arithmetic sequence with 3 as the common difference.

We could write this sequence as:

$$8, \quad 8 + 3, \quad 8 + 3 + 3, \quad 8 + 3 + 3 + 3, \quad 8 + 3 + 3 + 3 + 3, \quad \dots$$

or

$$8, \quad 8 + 3, \quad 8 + (2 \times 3), \quad 8 + (3 \times 3), \quad 8 + (4 \times 3), \quad \dots$$



Arithmetic sequences

If we call the first term of an arithmetic sequence a and the common difference d we can write a general arithmetic sequence as:

$$a, \quad a + d, \quad a + 2d, \quad a + 3d, \quad a + 4d, \dots$$

The n^{th} term of an arithmetic sequence with first term a and common difference d is

$$a + (n - 1)d$$

Also:

The inductive definition of an arithmetic sequence with first term a and common difference d is

$$u_1 = a, \quad u_{n+1} = u_n + d$$



Arithmetic sequences

What is the formula for the n^{th} term of the sequence 10, 7, 4, 1, $-2 \dots$?

This is an arithmetic sequence with first term $a = 10$ and common difference $d = -3$.

The n^{th} term is given by $a + (n - 1)d$ so:

$$\begin{aligned}u_n &= 10 - 3(n - 1) \\&= 10 - 3n + 3 \\&= 13 - 3n\end{aligned}$$

Let's check this formula for the first few terms in the sequence:

$$u_1 = 13 - 3 \times 1 = 10$$

$$u_2 = 13 - 3 \times 2 = 7$$

$$u_3 = 13 - 3 \times 3 = 4$$



Arithmetic sequences

Find the number of terms in the finite arithmetic sequence $-7, -1, 5, \dots, 71$.

This is an arithmetic sequence with first term $a = -7$ and common difference $d = 6$.

The n^{th} term is given by $a + (n - 1)d$ so:

$$\begin{aligned}u_n &= -7 + 6(n - 1) \\&= -7 + 6n - 6 \\&= 6n - 13\end{aligned}$$

We can find the value of n for the last term by solving:

$$\begin{aligned}6n - 13 &= 71 \\6n &= 84 \\n &= 14\end{aligned}$$

So, there are 14 terms in the sequence.



Arithmetic sequences

The 4th term in an arithmetic sequence is 12 and the 20th term is 92. What is the formula for the n^{th} term of this sequence?

Using the 4th term: $a + 3d = 12$

Using the 20th term: $a + 19d = 92$

Subtracting the first equation from the second equation gives:

$$16d = 80$$

$$d = 5$$

Substitute this into the first equation:

$$a + 15 = 12$$

$$a = -3$$

The n^{th} term of an arithmetic sequence with $a = -3$ and $d = 5$ is:

$$u_n = -3 + 5(n - 1)$$

$$= -3 + 5n - 5$$

$$= 5n - 8$$



Series

The sum of all the terms of a sequence is called a **series**.

For example:

1, 3, 5, 7, 9, ... is a sequence

while: $1 + 3 + 5 + 7 + 9 + \dots$ is a series.

When the difference between each term in a series is constant, as in this example, the series is called an **arithmetic series** or **arithmetic progression (AP)** for short).

The sum of a series containing n terms is often denoted by S_n , so for the series given above we could write:

$$\begin{aligned} S_5 &= 1 + 3 + 5 + 7 + 9 \\ &= 25 \end{aligned}$$

When n is large, a more systematic approach for calculating the sum of a given number of terms is required.



Gauss' method

It is said that when the famous mathematician Karl Friedrich Gauss was a young boy at school, his teacher asked the class to add together every whole number from one to a hundred.



The teacher expected this activity to keep the class occupied for some time and so he was amazed when Gauss put up his hand and gave the answer, 5050, almost immediately!



Gauss' method

Gauss worked the answer out by noticing that you can quickly add together consecutive numbers by writing the numbers once in order and once in reverse order and adding them together.

So to add the numbers from 1 to 100:

$$S = 1 + 2 + 3 + 4 + 5 + \dots + 98 + 99 + 100$$

$$S = 100 + 99 + 98 + 97 + 96 + \dots + 3 + 2 + 1$$

$$2S = 101 + 101 + 101 + 101 + 101 + \dots + 101 + 101 + 101$$

So:

$$\begin{aligned} 2S &= 100 \times 101 \\ &= 10\ 100 \end{aligned}$$

$$S = 5050$$



The sum of the first n natural numbers

To find the sum of the first n natural numbers we can generalize Gauss' method as follows.

Write the sum of the first n natural numbers as:

$$S = 1 + 2 + 3 + \dots + (n-2) + (n-1) + n$$

$$S = n + (n-1) + (n-2) + \dots + 3 + 2 + 1$$

$$2S = (n+1) + (n+1) + (n+1) + \dots + (n+1) + (n+1) + (n+1)$$

This gives us:

$$2S = n(n+1)$$

So:

The sum of the first n natural numbers is given by

$$\frac{1}{2}n(n+1)$$



The sum of the first n natural numbers

What is the sum of the first 30 natural numbers?

$$1 + 2 + 3 + \dots + 30 = \frac{1}{2} \times 30 \times 31$$
$$= 465$$

What is the sum of the natural numbers from 21 to 30?

$$21 + 22 + 23 + \dots + 30 = (1 + 2 + \dots + 30) - (1 + 2 + \dots + 20)$$
$$= 465 - \frac{1}{2} \times 20 \times 21$$
$$= 465 - 210$$
$$= 255$$



The sum of an arithmetic series

Gauss' method can be applied to any arithmetic series of the general form

$$a + (a + d) + (a + 2d) + (a + 3d) + \dots + (a + (n - 1)d)$$

where a is the first term in the series, d is the common difference and n is the number of terms.

Let's call the last term l so that:

$$l = (a + (n - 1)d)$$

The sum of the first n terms can now be written as:

$$S_n = a + (a + d) + (a + 2d) + \dots + (l - 2d) + (l - d) + l$$

$$S_n = l + (l - d) + (l - 2d) + \dots + (a + 2d) + (a + d) + a$$

$$2S_n = (a + l) + (a + l) + (a + l) + \dots + (a + l) + (a + l) + (a + l)$$



The sum of an arithmetic series

This gives us:

$$2S_n = n(a + l)$$

So:

The sum of the first n terms in an arithmetic series is

$$S_n = \frac{n}{2}(a + l)$$

where a is the first term and l is the last.

If the last term is not known this formula can be written in terms of a and n by substituting $(a + (n - 1)d)$ for l in the above.

An alternative formula for the sum of an arithmetic series is then:

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$



The sum of an arithmetic series

Find the sum of the first 20 terms of the arithmetic series
 $5 + 11 + 17 + 23 + \dots$

We don't know the last term so we can use:

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

with $a = 5$, $d = 6$ and $n = 20$.

$$S_{20} = \frac{20}{2}(2 \times 5 + 19 \times 6)$$

$$S_{20} = 10(10 + 114)$$

$$= 1240$$



Using Σ notation

When working with series, the Greek symbol Σ (the capital letter sigma) is used to mean 'the sum of'.

For example:

... and this is the last value of r .

$$\sum_{r=1}^n u_r$$

This is the first value of r ...

represents a finite series containing n terms:

$$u_1 + u_2 + u_3 + \dots + u_n$$

The terms in the series are obtained by substituting 1, 2, 3, ..., n in turn for r in u_r .



Using Σ notation

For example, suppose we want to find the sum of the first 4 terms of the series whose n^{th} term is of the form $3n - 1$.

We can write:

$$\begin{aligned}\sum_{r=1}^4 3r - 1 &= (3 \times 1 - 1) + (3 \times 2 - 1) + (3 \times 3 - 1) + (3 \times 4 - 1) \\ &= 2 + 5 + 8 + 11\end{aligned}$$

The initial value of r doesn't have to be 1. For example:

$$\sum_{r=3}^8 r^2 = 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2$$

Infinite series are given by writing an ∞ symbol above the Σ .
For example:

$$\sum_{r=1}^{\infty} \frac{1}{r+1} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$





Usii

. . .



Using Σ notation

$$\text{Evaluate } \sum_{r=2}^{15} 25 - 2r$$

Substituting $r = 2, 3, 4, \dots, 15$ into $25 - 2r$ gives us the arithmetic series $21 + 19 + 17 + 15 + \dots + -5$.

There are 14 terms in this sequence because both $r = 2$ and $r = 15$ are included.

We can evaluate this by putting $a = 21$, $l = -5$ and $n = 14$ into the formula

$$S_n = \frac{n}{2}(a + l)$$

So:

$$S_{14} = \frac{14}{2}(21 + -5)$$
$$= 112$$



- Sequences
- The formula for the n^{th} term
- Recurrence relations
- Arithmetic sequences
- Arithmetic series
- The sum of the first n natural numbers
- The sum of an arithmetic series
- Using Σ notation
- **Examination-style questions**



Exa

The sum of the first 3 terms of an arithmetic series is 21 and the sum of the next three terms is 66.

- a) Find the value of the first term and the common difference.
- b) Write an expression for the n^{th} term of the series u_n .
- c) Find the sum of the first 10 terms.

a) The sum of the first 3 terms can be written as:

$$a + (a + d) + (a + 2d) = 3a + 3d$$

So

$$3a + 3d = 21$$

$$a + d = 7 \quad \textcircled{1}$$

The sum of the next 3 terms can be written as:

$$(a + 3d) + (a + 4d) + (a + 5d) = 3a + 12d$$

So

$$3a + 12d = 66$$

$$a + 4d = 22 \quad \textcircled{2}$$



Examination-style question

② – ① :

$$3d = 15$$

$$d = 5$$

$$a = 2$$

b) In general, for an arithmetic series $u_n = a + (n - 1)d$ so

$$u_n = 2 + 5(n - 1)$$

$$= 5n - 3$$

c)

$$u_{10} = (5 \times 10) - 3$$

$$= 47$$

Now using the formula $S_n = \frac{n}{2}(a + l)$ with $a = 2$ and $l = 47$:

$$S_{10} = \frac{10}{2}(2 + 47)$$

$$= 245$$

Definition: A set is an unordered collection of objects

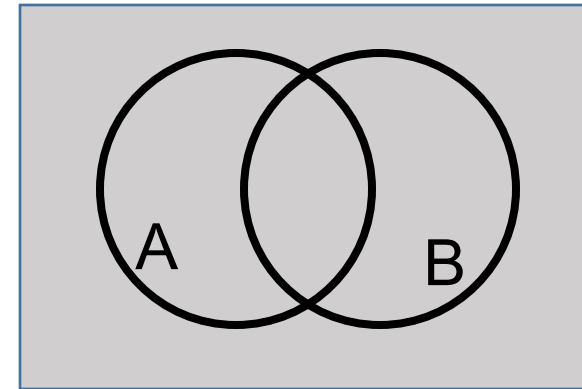
Cartesian Product : $A \times B$

$$A \times B = \{ (a, b) \mid a \in A \wedge b \in B \}$$

De Morgan's Laws

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$



Proof technique:

To show $C = D$ show

$x \in C \rightarrow x \in D$ and

$x \in D \rightarrow x \in C$

Russell's Paradox

$$S = \{ x \mid x \notin x \}$$

Counting Rules

Product Rule: If there are n_1 choices for the first item and n_2 choices for the second item, then there are $n_1 n_2$ choices for the two items

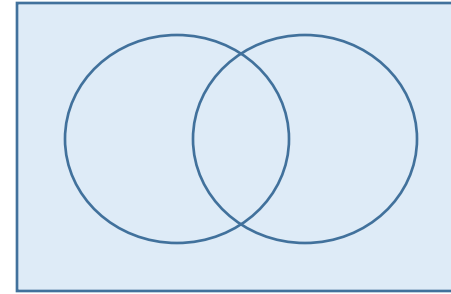
Sum Rule: If there are n_1 choices of an element from S_1 and n_2 choices of an element from S_2 and $S_1 \cap S_2$ is empty, then there are $n_1 + n_2$ choices of an element from $S_1 \cup S_2$

Important cases of the Product Rule

- Cartesian product
 - $|A_1 \times A_2 \times \dots \times A_n| = |A_1| |A_2| \dots |A_n|$
- Subsets of a set S
 - $|P(S)| = 2^{|S|}$
- Strings of length n over Σ
 - $|\Sigma^n| = |\Sigma|^n$

Inclusion-Exclusion Principle

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$$

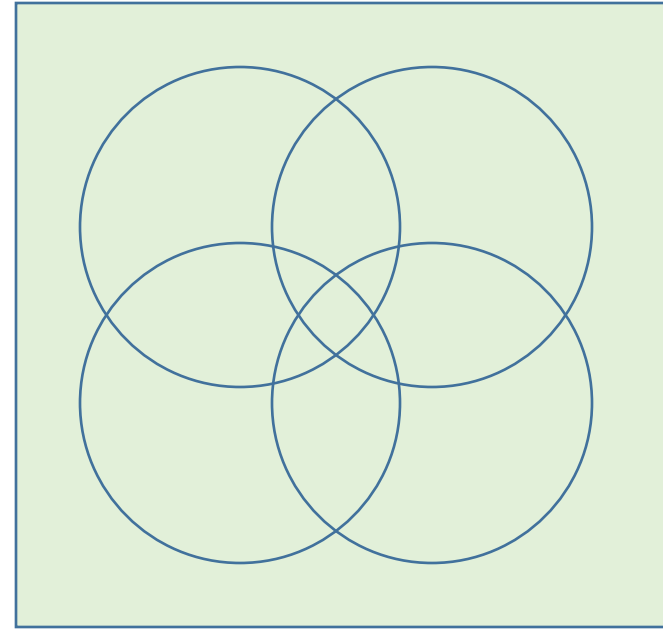
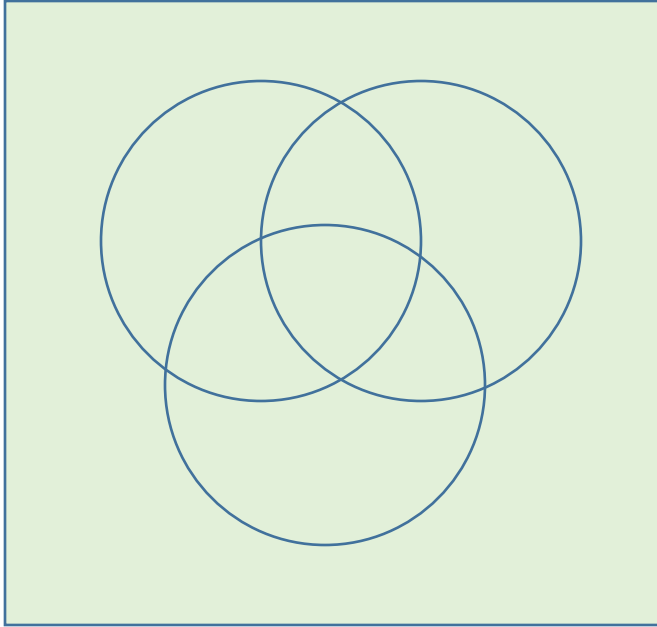


- How many binary strings of length 9 start with 00 or end with 11

Inclusion-Exclusion

- A class has of 40 students has 20 CS majors, 15 Math majors. 5 of these students are dual majors. How many students in the class are neither math, nor CS majors?

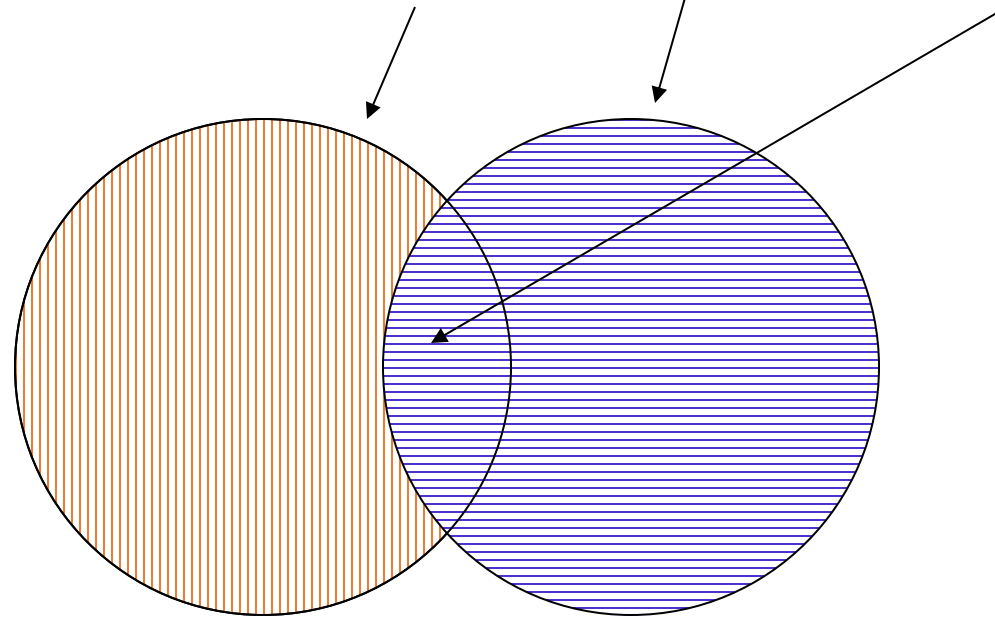
Generalizing Inclusion Exclusion



§6.5 The principle of Inclusion-Exclusion

If A , B and C are finite sets *then*

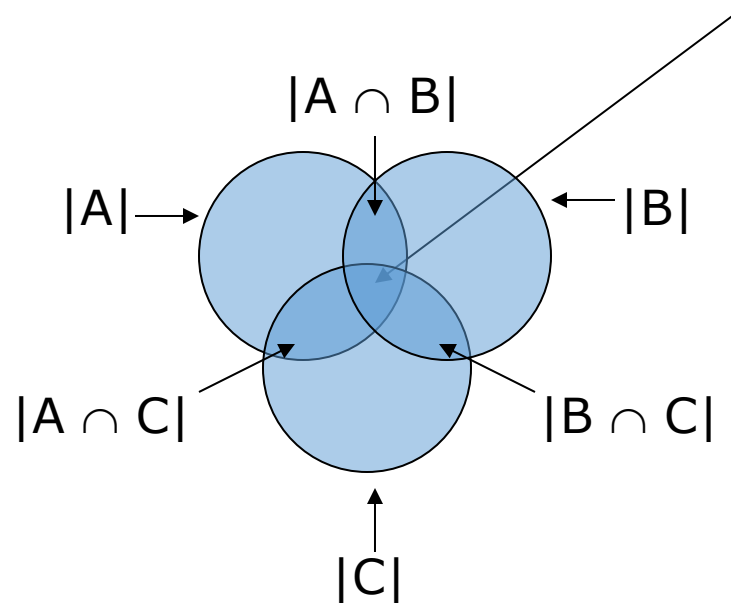
$$|A \cup B| = |A| + |B| - |A \cap B|$$



Continue...

- If A, B and C are finite sets then

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| - |A \cap B \cap C|$$



General Inclusion-Exclusion

- **Theorem 1:** Let $A_1, \dots, A_n$ be finite sets.

Then $|A_1 \cup A_2 \cup \dots \cup A_n| =$

$$\sum_{1 \leq i \leq n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|$$

Combinatorics

- Count the number of ways to put things together into various combinations.
e.g. If a password is 6-8 letters and/or digits, how many passwords can there be?
- Two main rules:
 - Sum rule
 - Product rule

Sum Rule

- Let us consider two tasks:
 - m is the number of ways to do **task 1**
 - n is the number of ways to do **task 2**
 - Tasks are independent of each other, i.e.,
 - Performing **task 1** does not accomplish **task 2** and vice versa.
- Sum rule: the number of ways that “**either** task 1 **or** task 2 can be done, but **not both**”, is $m+n$.
- Generalizes to multiple tasks ...

Example

- A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects respectively. How many possible projects are there to choose from?

Set Theoretic Version

- If A is the set of ways to do task 1, and B the set of ways to do task 2, and if A and B are disjoint, then:

“the ways to do either task 1 or 2 are

$A \cup B$, and $|A \cup B| = |A| + |B|$ ”

Product Rule

- Let us consider two tasks:
 - m is the number of ways to do **task 1**
 - n is the number of ways to do **task 2**
 - Tasks are independent of each other, i.e.,
 - Performing task 1 does not accomplish task 2 and vice versa.
- Product rule: the number of ways that “**both** tasks 1 and 2 can be done” in mn .
- Generalizes to multiple tasks ...

Example

- The chairs of an auditorium are to be labeled with a letter and a positive integer not to exceed 100. What is the largest number of chairs that can be labeled differently?

Set Theoretic Version

- If A is the set of ways to do task 1, and B the set of ways to do task 2, and if A and B are disjoint, then:
- The ways to do both task 1 and 2 can be represented as $A \times B$, and $|A \times B| = |A| \cdot |B|$

More Examples

- How many different bit strings are there of length seven?

More Examples

- Suppose that either a member of the CS faculty or a student who is a CS major can be on a university committee. How many different choices are there if there are 37 CS faculty and 83 CS majors ?

More Examples

- How many different license plates are available if each plate contains a sequence of three letters followed by three digits?

More Examples

- What is the number of different subsets of a finite set S ?

Example Using Both Rules

- Each user on a computer system has a password, which is six to eight characters long where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Inclusion-Exclusion Principle (relates to the “sum rule”)

- Suppose that $k \leq m$ of the ways of doing task 1 also simultaneously accomplishes task 2. (And thus are also ways of doing task 2.)
- Then the number of ways to accomplish “Do either task 1 or task 2” is $m+n-k$.
- **Set theory:** If A and B are not disjoint, then $|A \cup B| = |A| + |B| - |A \cap B|$.

Pigeonhole Principle

- If $k+1$ objects are assigned to k places, then **at least** 1 place must be assigned ≥ 2 objects.
- In terms of the assignment function:
If $f:A \rightarrow B$ and $|A| \geq |B| + 1$, then some element of B has ≥ 2 pre-images under f .
i.e., f is **not** one-to-one.

Example

- How many students must be in class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points?

Generalized Pigeonhole Principle

- If $N \geq k+1$ objects are assigned to k places, then **at least** one place must be assigned at **least** $\lceil N/k \rceil$ objects.
- *e.g.*, there are $N=280$ students in this class. There are $k=52$ weeks in the year.
 - Therefore, there must be at **least** 1 week during which **at least** $\lceil 280/52 \rceil = \lceil 5.38 \rceil = 6$ students in the class have a birthday.

Proof of G.P.P.

- By contradiction. Suppose **every** place has $< \lceil N/k \rceil$ objects, thus $\leq \lceil N/k \rceil - 1$.
- Then the total number of objects is at most

- So, there are less than N objects, which contradicts our assumption of N objects! $\square$

G.P.P. Example

- Given: There are 280 students in the class. Without knowing anybody's birthday, what is the **largest** value of n for which we can prove that at **least** n students must have been born in the same month?
- Answer:

$$\lceil 280/12 \rceil = \lceil 23.3 \rceil = 24$$

More Examples

- What is the **minimum** number of students required in a discrete math class to be sure that at **least** six will receive the same grade, if there are five possible grades, A, B, C, D, and F?

Permutations

- A *permutation* of a set S of objects is an ordered arrangement of the elements of S where each element appears only once:

e.g., 1 2 3, 2 1 3, 3 1 2

- An ordered arrangement of r distinct elements of S is called an *r-permutation*.
- The number of r -permutations of a set S with $n=|S|$ elements is

$$P(n,r) = n(n-1)\dots(n-r+1) = n!/(n-r)!$$

Example

- How many ways are there to select a third-prize winner from 100 different people who have entered a contest?

More Examples

- A terrorist has planted an armed nuclear bomb in your city, and it is your job to disable it by cutting wires to the trigger device.
- There are 10 wires to the device.
- If you cut exactly the right three wires, in exactly the right order, you will disable the bomb, otherwise it will explode!
- If the wires all look the same, what are your chances of survival?

$P(10,3) = 10 \cdot 9 \cdot 8 = 720$,
so there is a 1 in 720 chance
that you'll survive!

More Examples

- How many permutations of the letters ABCDEFG contain the string ABC?

Combinations

- The number of ways of choosing r elements from S (order does not matter).

$$S=\{1,2,3\}$$

e.g., 1 2 , 1 3, 2 3

- The number of r -combinations $C(n,r)$ of a set with $n=|S|$ elements is

$$C(n,r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Combinations vs Permutations

- Essentially unordered permutations ...

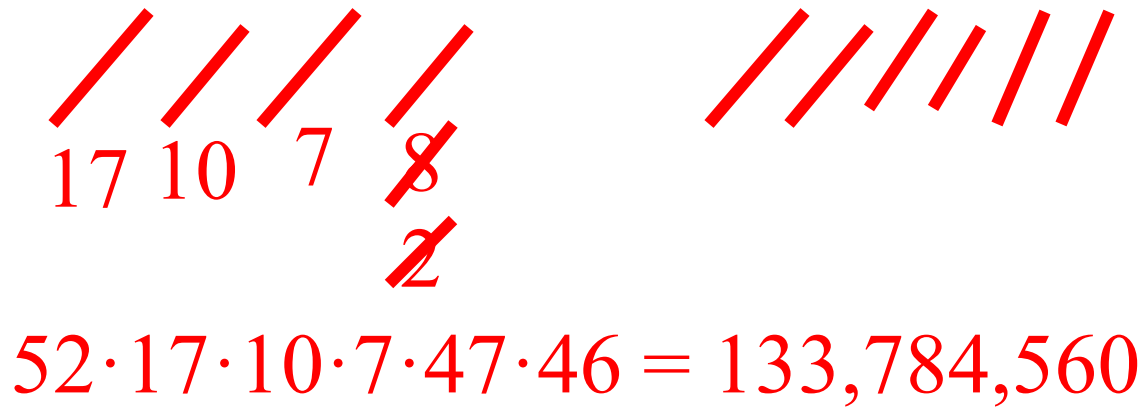
$$P(n, r) = C(n, r)P(r, r)$$

$$C(n, r) = \binom{n}{r} = \frac{P(n, r)}{P(r, r)} = \frac{n!/(n-r)!}{r!} = \frac{n!}{r!(n-r)!}$$

- Note that $C(n, r) = C(n, n-r)$

Combination Example

- How many distinct 7-card hands can be drawn from a standard 52-card deck?
 - The order of cards in a hand doesn't matter.
- Answer $C(52,7) = P(52,7)/P(7,7)$
 $= 52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 \cdot 47 \cdot 46 / 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$


$$52 \cdot 17 \cdot 10 \cdot 7 \cdot 47 \cdot 46 = 133,784,560$$

More Examples

- How many ways are there to select a committee to develop a discrete mathematics course if the committee is to consist of 3 faculty members from the Math department and 4 from the CS department, if there are 9 faculty members from Math and 11 from CS?

Generalized Permutations and Combinations

- How to solve counting problems where elements may be used more than once?
- How to solve counting problems in which some elements are not distinguishable?
- How to solve problems involving counting the ways we to place distinguishable elements in distinguishable boxes?

Permutations with Repetition

- The number of r -permutations of a set of n objects with **repetition** allowed is n^r
- Example: How many strings of length n can be formed from the English alphabet?

Combinations with Repetition

- The number of r -combinations from a set with n elements when **repetition** of elements is allowed are $C(n+r-1, r)$

Combinations with Repetition

Example: How many ways are there to select 5 bills from a cash box containing \$1 bills, \$2 bills, \$5 bills, \$10 bills, \$20 bills, \$50 bills, and \$100 bills? Assume that the order in which bills are chosen does not matter and there are at least 5 bills of each type.

Permutations and Combinations with and without Repetition

TABLE 1 Combinations and Permutations with and without Repetition.

<i>Type</i>	<i>Repetition Allowed?</i>	<i>Formula</i>
r -permutations	No	$\frac{n!}{(n-r)!}$
r -combinations	No	$\frac{n!}{r!(n-r)!}$
r -permutations	Yes	n^r
r -combinations	Yes	$\frac{(n+r-1)!}{r!(n-1)!}$

Permutations with non-distinguishable objects

- The number of different permutations of n objects, where there are n_1 non-distinguishable objects of type 1, n_2 non-distinguishable objects of type 2, ..., and n_k non-distinguishable objects of type k, is

$$\frac{n!}{n_1! n_2! \dots n_k!}$$

i.e., $C(n, n_1) C(n - n_1, n_2) \dots C(n - n_1 - n_2 - \dots - n_{k-1}, n_k)$

$$n_1 + n_2 + \dots + n_k = n$$

$$n_1 + n_2 + \dots + n_k = n$$

Binomial Coefficient Identities from the Binomial Theorem

$$\sum_{k=0}^n \binom{n}{k} x^{n-k} y^k = (x + y)^n$$

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

$$\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$$

$$\sum_{k=0}^n 2^k \binom{n}{k} = 3^n$$

Definition of Relations

Let A and B be sets,
A **binary relation from A to B** is a subset of $A \times B$

Let A be a set,
A **binary relation on A** is a subset of $A \times A$

Combining Relations

Let R be a relation from A to B

Let S be a relation from B to C

The composite of R and S , $S \circ R$ is the relation from A to C defined

$$S \circ R = \{(a, c) \mid \exists b \text{ such that } (a,b) \in R \text{ and } (b,c) \in S\}$$

Powers of a Relation

$$R^2 = R \circ R = \{(a, c) \mid \exists b \text{ such that } (a,b) \in R \text{ and } (b,c) \in R\}$$

$$R^0 = \{(a,a) \mid a \in A\}$$

$$R^1 = R$$

$$R^{n+1} = R^n \circ R$$

n-ary relations

Let $A_1, A_2, \dots, A_n$ be sets. An n -ary relation on these sets is a subset of $A_1 \times A_2 \times \dots \times A_n$.

Relational databases

Student_Name	ID_Number	Major	GPA
Knuth	328012098	CS	4.00
Von Neuman	481080220	CS	3.78
Von Neuman	481080220	Mathematics	3.78
Russell	238082388	Philosophy	3.85
Einstein	238001920	Physics	2.11
Newton	1727017	Mathematics	3.61
Karp	348882811	CS	3.98
Newton	1727017	Physics	3.61
Bernoulli	2921938	Mathematics	3.21
Bernoulli	2921939	Mathematics	3.54

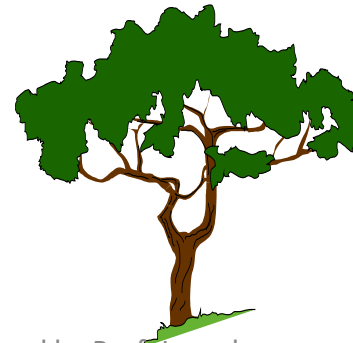
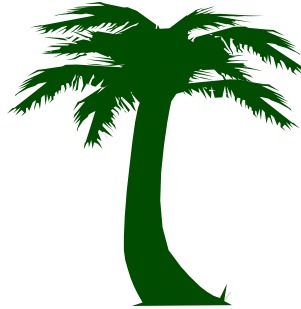
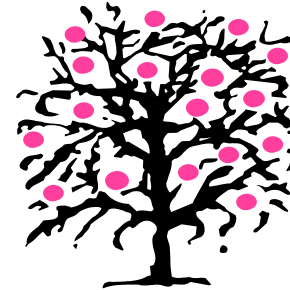
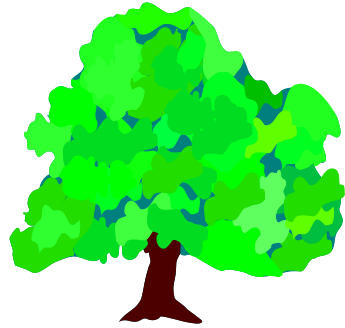
Alternate Approach

Student_ID	Name	GPA
328012098	Knuth	4.00
481080220	Von Neuman	3.78
238082388	Russell	3.85
238001920	Einstein	2.11
1727017	Newton	3.61
348882811	Karp	3.98
2921938	Bernoulli	3.21
2921939	Bernoulli	3.54

Student_ID	Major
328012098	CS
481080220	CS
481080220	Mathematics
238082388	Philosophy
238001920	Physics
1727017	Mathematics
348882811	CS
1727017	Physics
2921938	Mathematics
2921939	Mathematics

Section 10.1

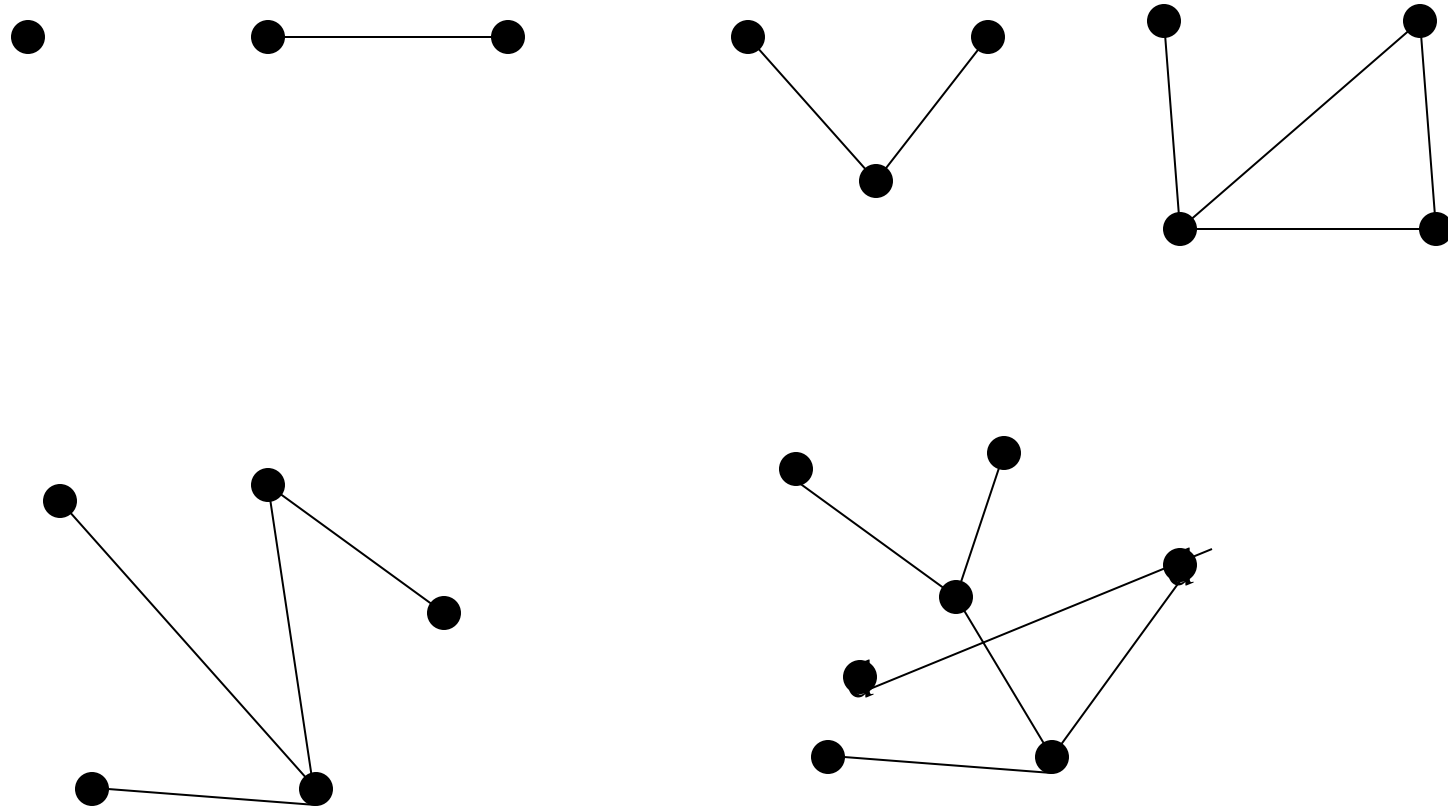
Introduction to Trees



Definition:

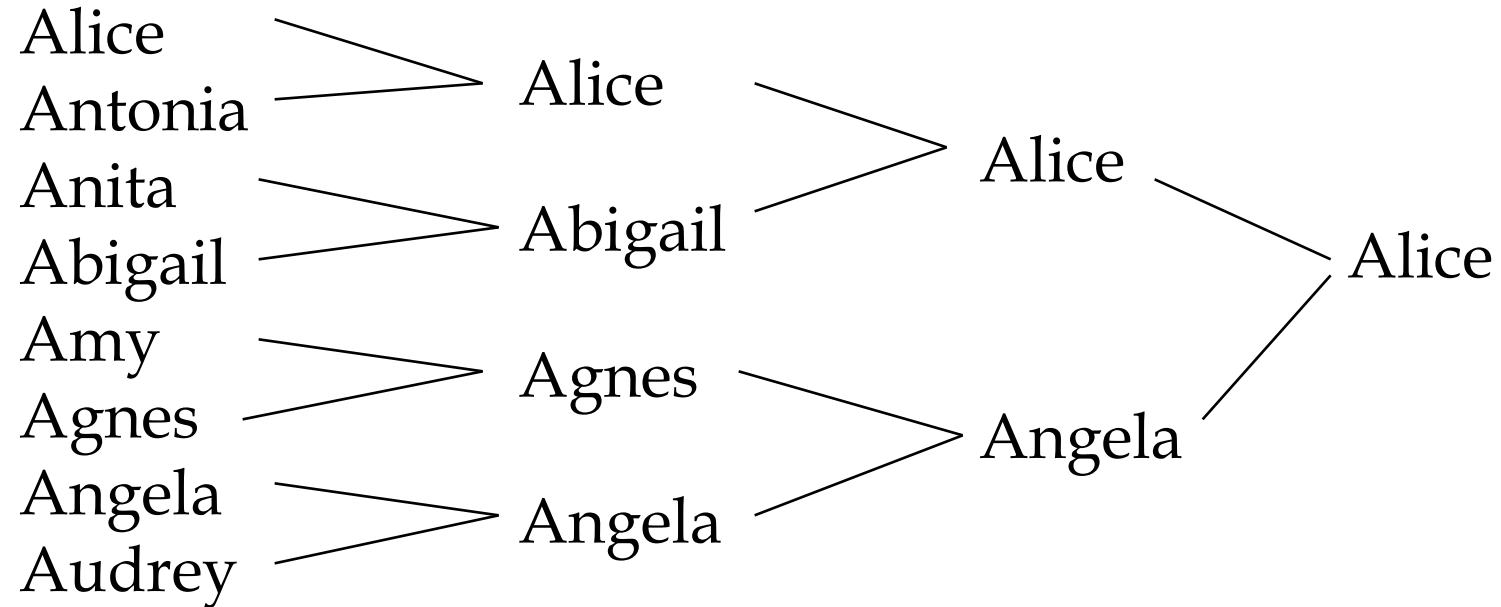
A *tree* is a connected undirected graph with no simple circuits.

Recall: A **circuit** is a path of length ≥ 1 that begins and ends at the same vertex.



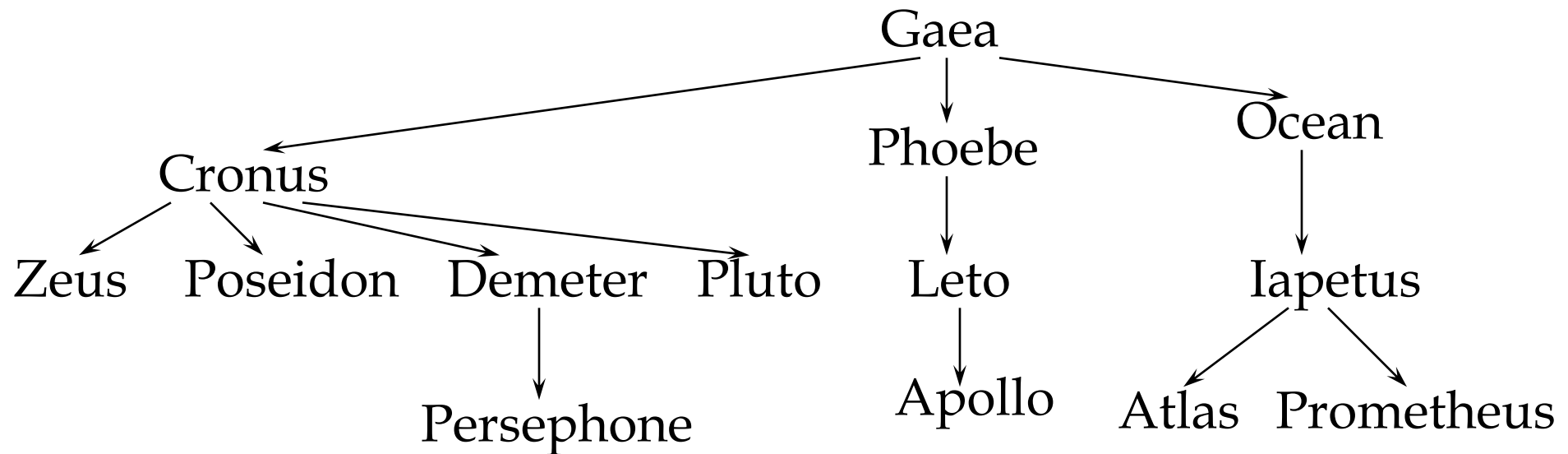
Tournament Trees

A common form of tree used in everyday life is the tournament tree, used to describe the outcome of a series of games, such as a tennis tournament.



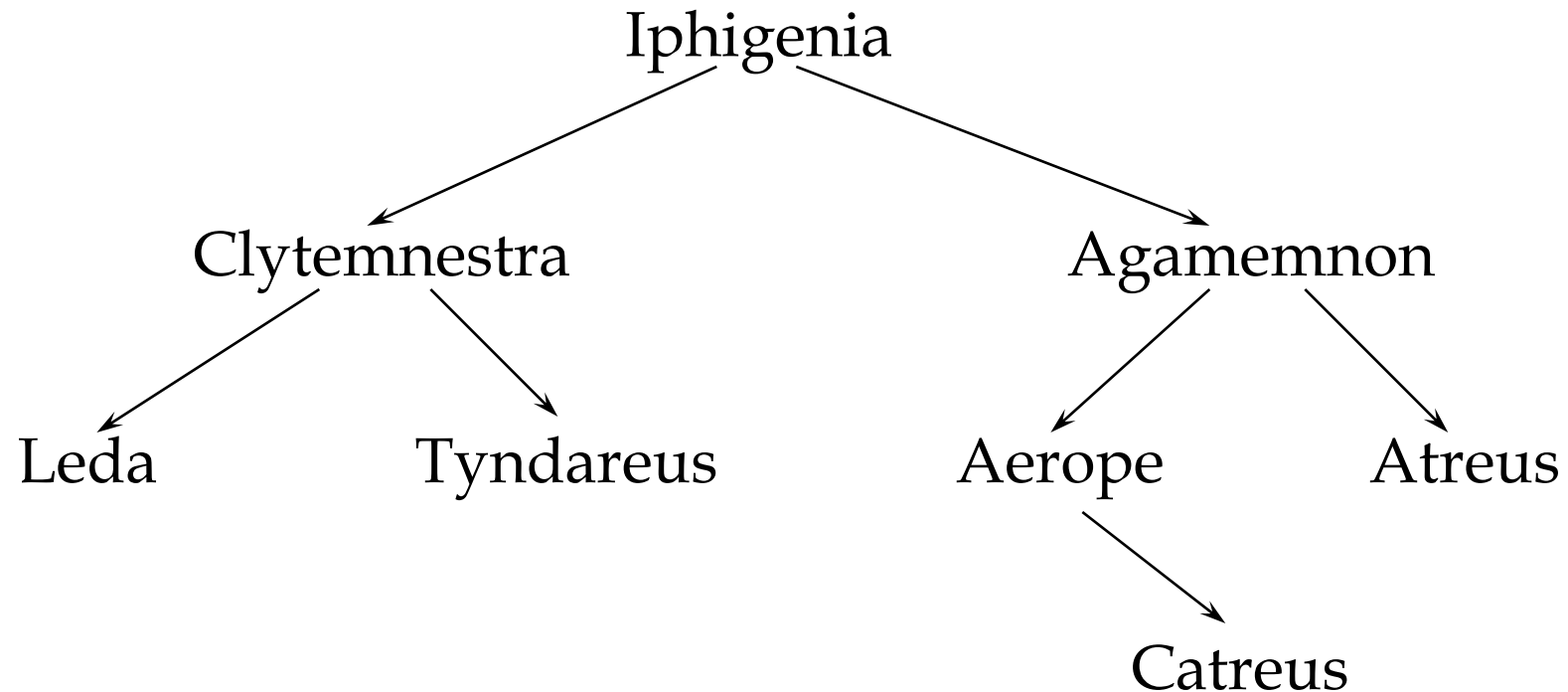
A Family Tree

Much of the tree terminology derives from family trees.



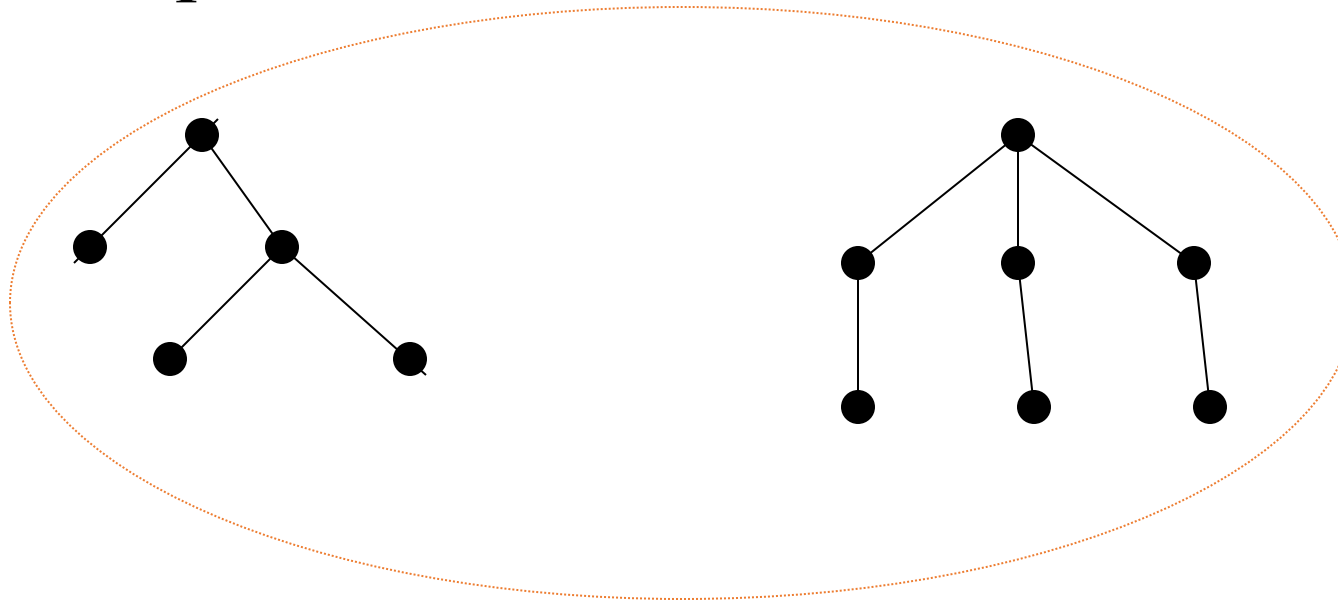
Ancestor Tree

An inverted family tree. Important point - it is a *binary* tree.



Forest

Graphs containing no simple circuits that are not connected, but each connected component is a tree.



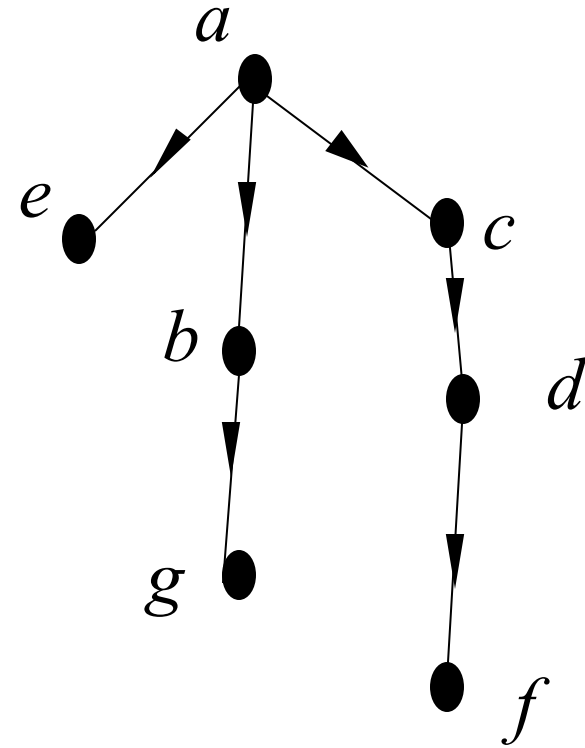
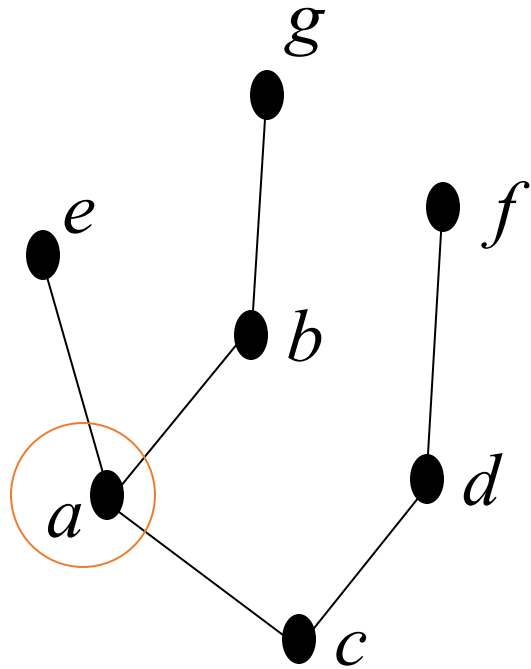
Theorem

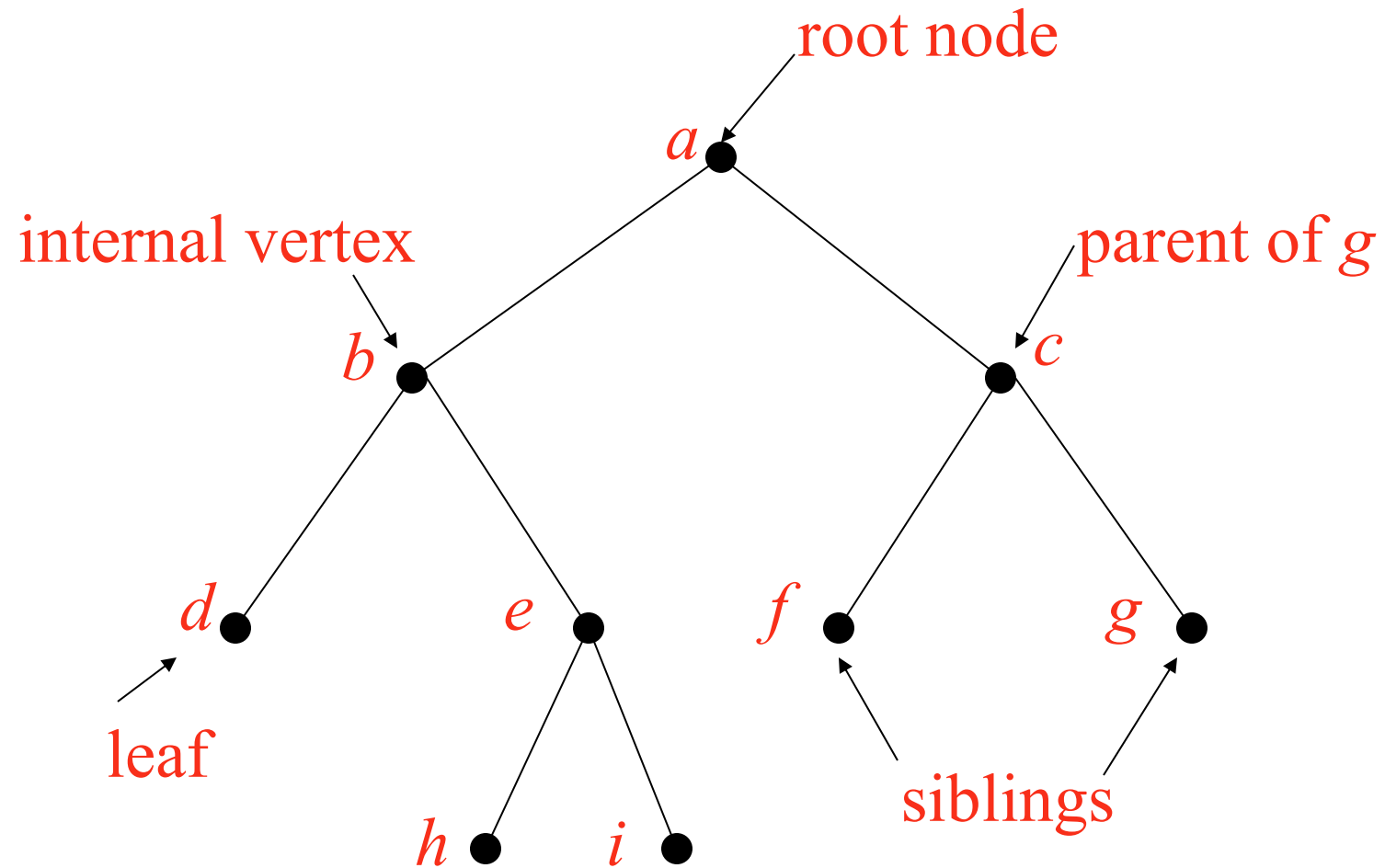
An undirected graph is a tree if and only if there is a unique simple path between any two of its vertices.

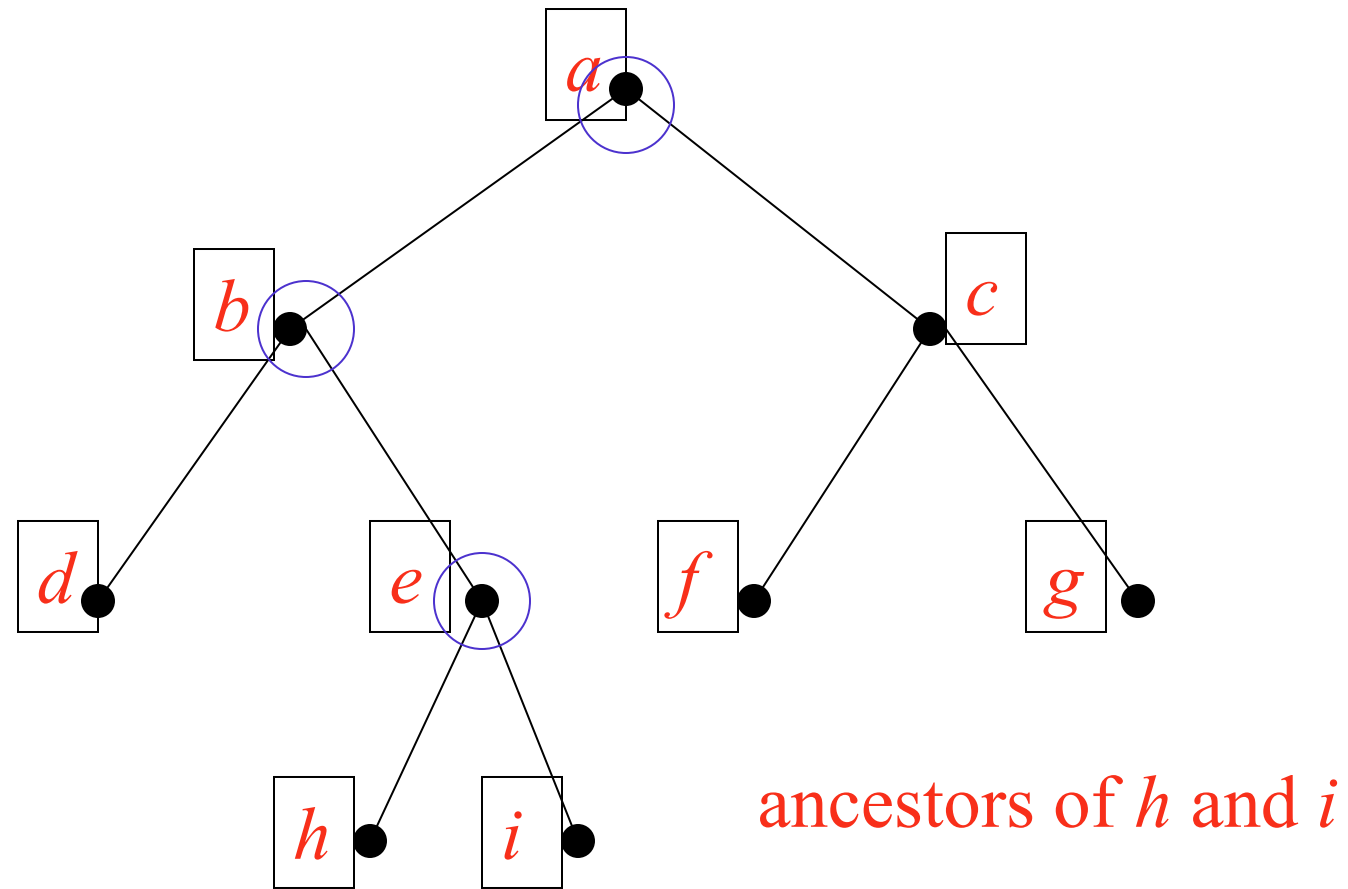
Rooted Trees

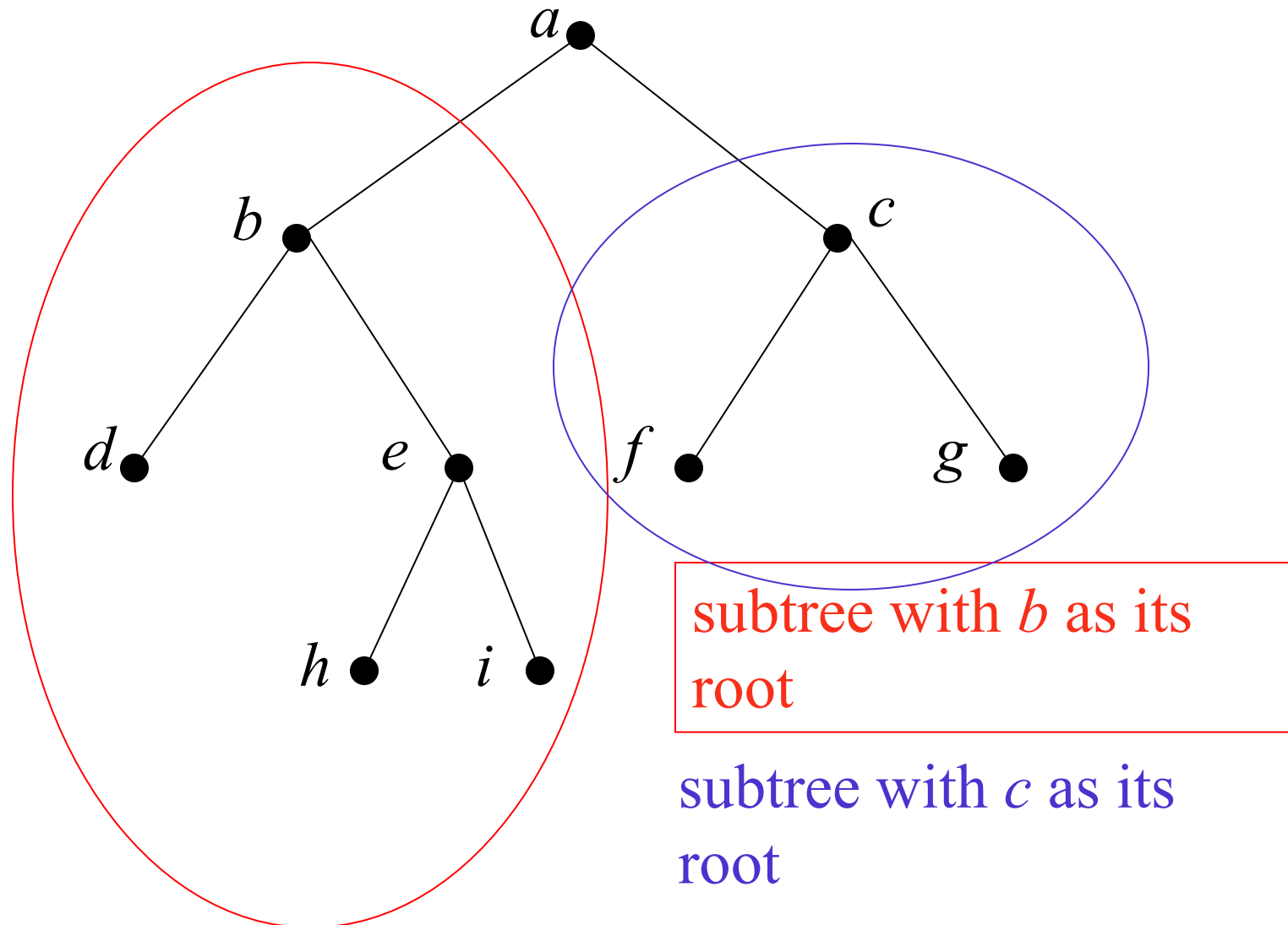
Once a vertex of a tree has been designated as the *root* of the tree, it is possible to assign direction to each of the edges.

Rooted Trees



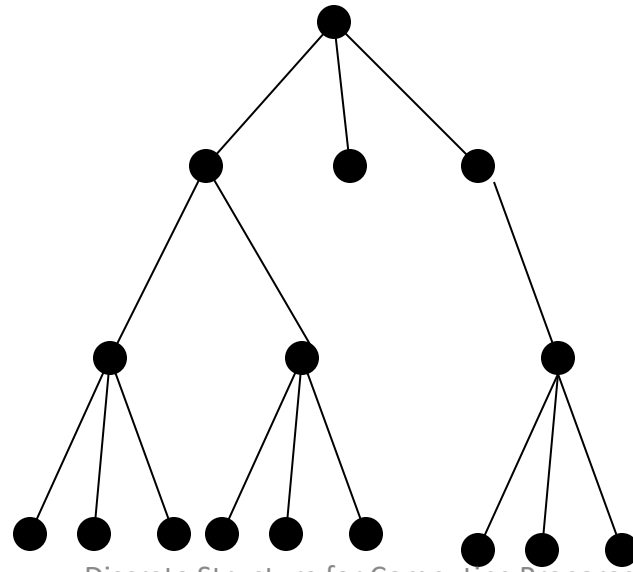
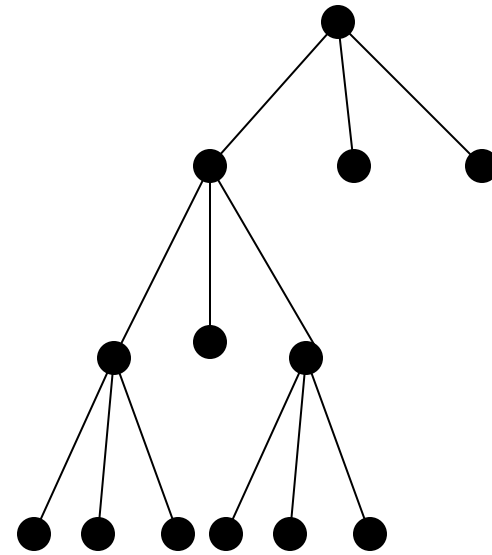
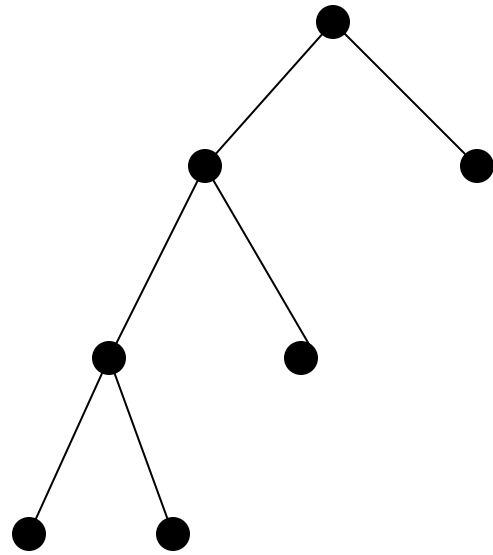






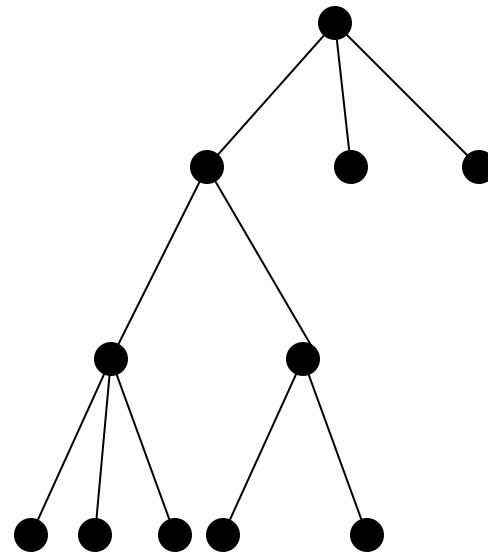
m -ary trees

A rooted tree is called an *m -ary tree* if every internal vertex has no more than m children. The tree is called a *full m -ary tree* if every internal vertex has exactly m children. An *m -ary tree* with $m=2$ is called a *binary tree*.



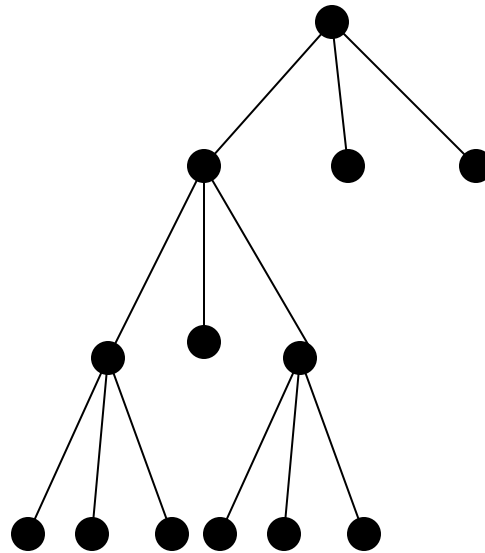
Properties of Trees

A tree with n vertices has $n-1$ edges.



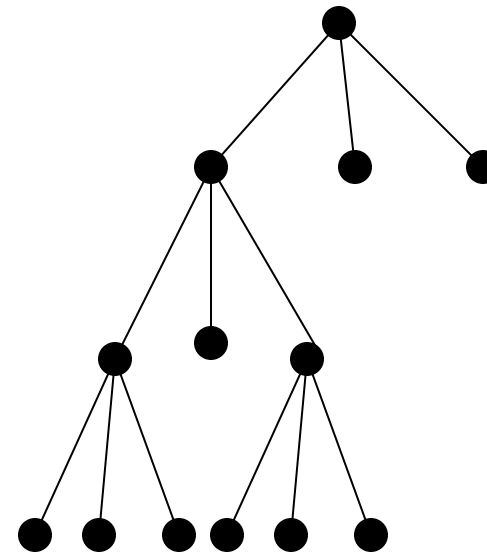
Properties of Trees

A full m -ary tree with i internal vertices contains $n = mi+1$ vertices.



Properties of Trees

The *height* of a rooted tree is the maximum of the levels of vertices.



Matrix representation

Relation R from $A=\{a_1, \dots a_p\}$ to $B=\{b_1, \dots b_q\}$

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R, \\ 0 & \text{if } (a_i, b_j) \notin R. \end{cases}$$

$\{(1, 1), (1, 2), (1, 4), (2,1), (2,3), (3,2), (3, 3) \}$

Graph Theory

- End of term material – limited chance for homework
- Cannot ask deep questions on the exam
- Graph theory is split across three classes
 - Algorithmic material is covered in other classes

Goals

- Understand the basic concept of a graph and associated terminology
- Model real world with graphs
 - Real world to formalism
- Elementary mathematical reasoning about graphs

Graph Theory

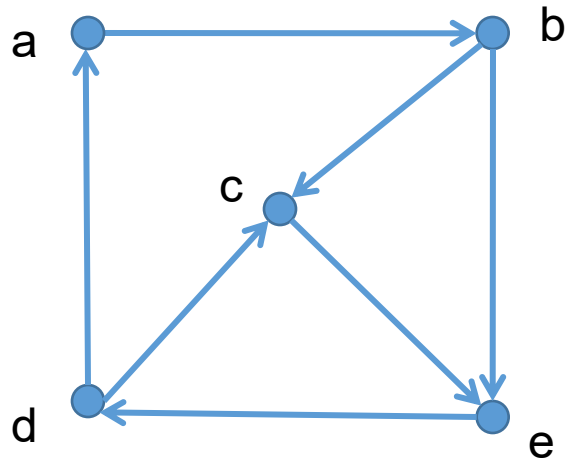
- Graph formalism
 - $G = (V, E)$
 - Vertices
 - Edges
- Directed Graph
 - Edges ordered pairs
- Undirected Graph
 - Edges sets of size two

Big Graphs

- Web Graph
 - Hyperlinks and pages
- Social Networks
 - Community Graph
 - Linked In, Face Book
 - Transactions
 - Ebay
 - Authorship
 - Erdos Number

Counting Paths

Let A be the Adjacency Matrix. What is A^2 ?



$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

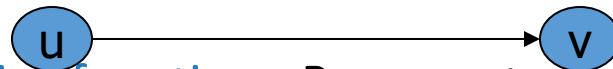
Definitions - Graph

A generalization of the simple concept of a set of dots, links, edges or arcs.

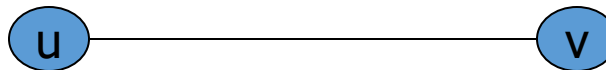
Representation: Graph $G = (V, E)$ consists set of vertices denoted by V , or by $V(G)$ and set of edges E , or $E(G)$

Definitions – Edge Type

Directed: Ordered pair of vertices. Represented as (u, v) directed from vertex u to v .

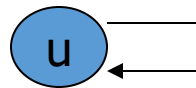


Undirected: Unordered pair of vertices. Represented as $\{u, v\}$. Disregards any sense of direction and treats both end vertices interchangeably.



Definitions – Edge Type

- **Loop:** A loop is an edge whose endpoints are equal i.e., an edge joining a vertex to it self is called a loop. Represented as $\{u, u\} = \{u\}$

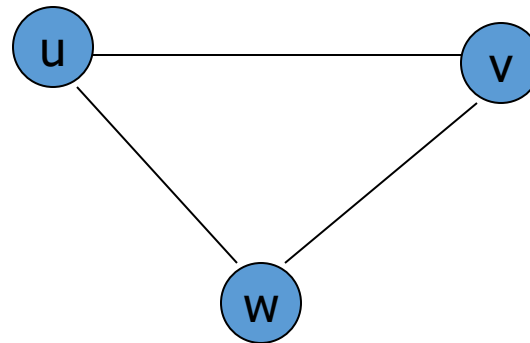


- **Multiple Edges:** Two or more edges joining the same pair of vertices.

Definitions – Graph Type

Simple (Undirected) Graph: consists of V , a nonempty set of vertices, and E , a set of unordered pairs of distinct elements of V called edges (undirected)

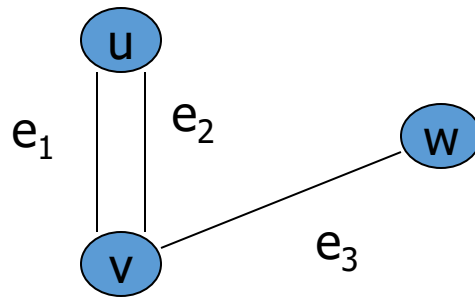
Representation Example: $G(V, E)$, $V = \{u, v, w\}$, $E = \{\{u, v\}, \{v, w\}, \{u, w\}\}$



Definitions – Graph Type

Multigraph: $G(V,E)$, consists of set of vertices V , set of Edges E and a function f from E to $\{\{u, v\} \mid u, v \in V, u \neq v\}$. The edges e_1 and e_2 are called multiple or parallel edges if $f(e_1) = f(e_2)$.

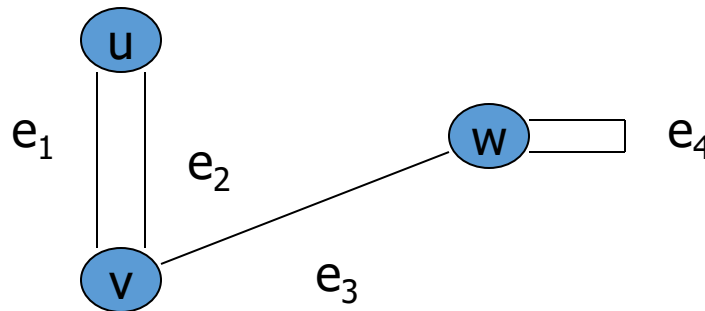
Representation Example: $V = \{u, v, w\}$, $E = \{e_1, e_2, e_3\}$



Definitions – Graph Type

Pseudograph: $G(V,E)$, consists of set of vertices V , set of Edges E and a function F from E to $\{\{u, v\} \mid u, v \in V\}$. Loops allowed in such a graph.

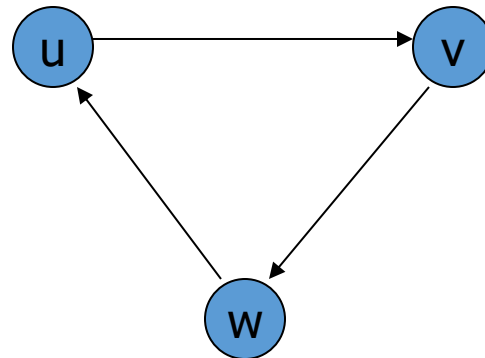
Representation Example: $V = \{u, v, w\}$, $E = \{e_1, e_2, e_3, e_4\}$



Definitions – Graph Type

Directed Graph: $G(V, E)$, set of vertices V , and set of Edges E , that are ordered pair of elements of V (directed edges)

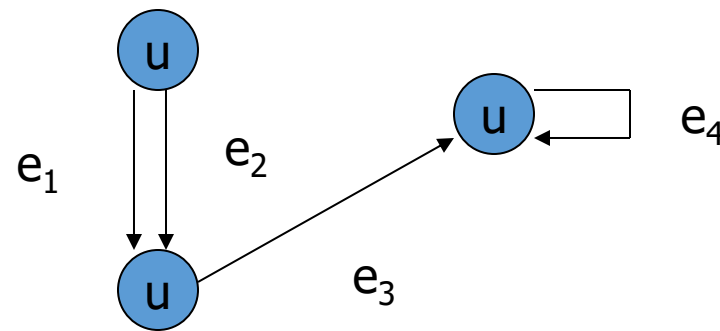
Representation Example: $G(V, E)$, $V = \{u, v, w\}$, $E = \{(u, v), (v, w), (w, u)\}$



Definitions – Graph Type

Directed Multigraph: $G(V,E)$, consists of set of vertices V , set of Edges E and a function f from E to $\{\{u, v\} \mid u, v \in V\}$. The edges e_1 and e_2 are multiple edges if $f(e_1) = f(e_2)$

Representation Example: $V = \{u, v, w\}$, $E = \{e_1, e_2, e_3, e_4\}$



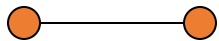
Simple graphs – special cases

- **Complete graph:** K_n , is the simple graph that contains exactly one edge between each pair of distinct vertices.

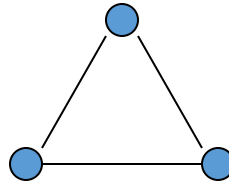
Representation Example: K_1 , K_2 , K_3 , K_4



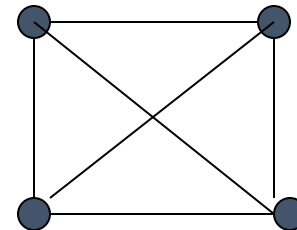
K_1



K_2



K_3

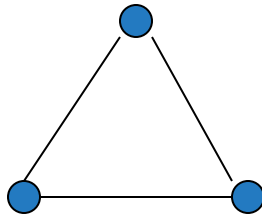


K_4

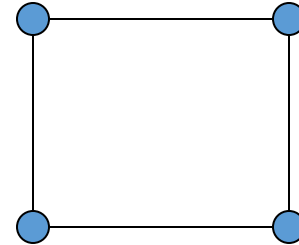
Simple graphs – special cases

- **Cycle:** C_n , $n \geq 3$ consists of n vertices $v_1, v_2, v_3 \dots v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \{v_3, v_4\} \dots \{v_{n-1}, v_n\}, \{v_n, v_1\}$

Representation Example: C_3, C_4



C_3



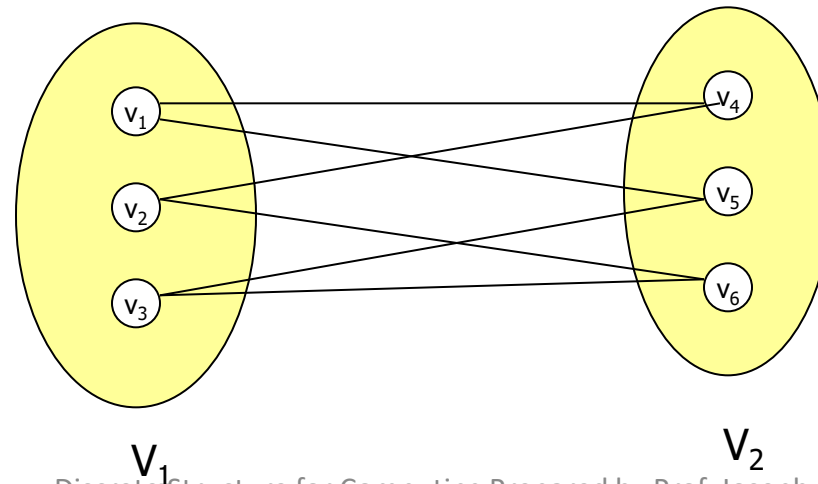
C_4

Bipartite graphs

- In a simple graph G , if V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 (so that no edge in G connects either two vertices in V_1 or two vertices in V_2)

Application example: Representing Relations

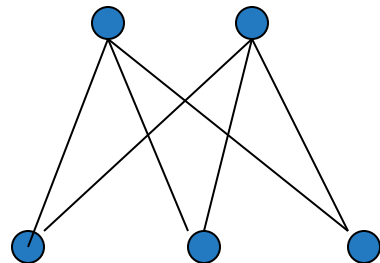
Representation example: $V_1 = \{v_1, v_2, v_3\}$ and $V_2 = \{v_4, v_5, v_6\}$,



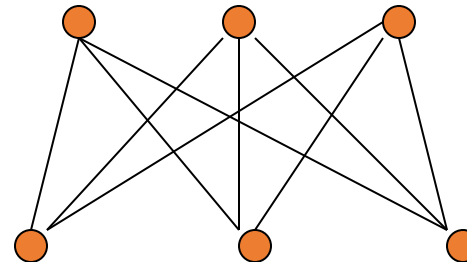
Complete Bipartite graphs

- $K_{m,n}$ is the graph that has its vertex set portioned into two subsets of m and n vertices, respectively. There is an edge between two vertices if and only if one vertex is in the first subset and the other vertex is in the second subset.

Representation example: $K_{2,3}$, $K_{3,3}$



$K_{2,3}$



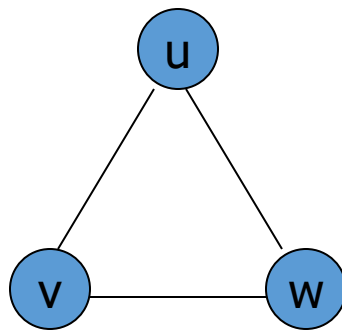
$K_{3,3}$

Subgraphs

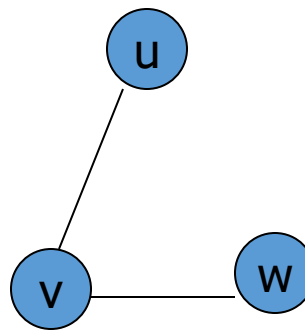
- A subgraph of a graph $G = (V, E)$ is a graph $H = (V', E')$ where V' is a subset of V and E' is a subset of E

Application example: solving sub-problems within a graph

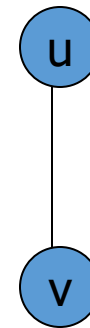
Representation example: $V = \{u, v, w\}$, $E = (\{u, v\}, \{v, w\}, \{w, u\})$, H_1, H_2



G



H_1

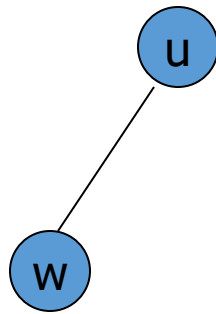


H_2

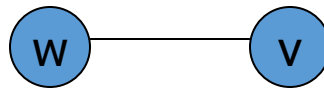
Subgraphs

- $G = G1 \cup G2$ wherein $E = E1 \cup E2$ and $V = V1 \cup V2$, G , $G1$ and $G2$ are simple graphs of G

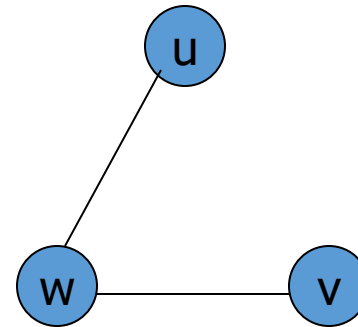
Representation example: $V1 = \{u, w\}$, $E1 = \{\{u, w\}\}$, $V2 = \{w, v\}$,
 $E2 = \{\{w, v\}\}$, $V = \{u, v, w\}$, $E = \{\{u, w\}, \{w, v\}\}$



G1



G2



G

Representation

- **Incidence (Matrix):** Most useful when information about edges is more desirable than information about vertices.
- **Adjacency (Matrix/List):** Most useful when information about the vertices is more desirable than information about the edges. These two representations are also most popular since information about the vertices is often more desirable than edges in most applications

Representation- Incidence Matrix

- $G = (V, E)$ be an undirected graph. Suppose that $v_1, v_2, v_3, \dots, v_n$ are the vertices and $e_1, e_2, \dots, e_m$ are the edges of G . Then the incidence matrix with respect to this ordering of V and E is the $n \times m$ matrix $M = [m_{ij}]$, where

Can also be used to represent :

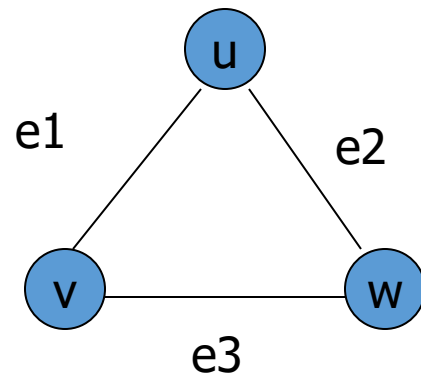
Multiple edges: by using columns with identical entries, since these edges are incident with the same pair of vertices

Loops: by using a column with exactly one entry equal to 1, corresponding to the vertex that is incident with the loop

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i \\ 0 & \text{otherwise} \end{cases}$$

Representation- Incidence Matrix

- Representation Example: $G = (V, E)$



	e_1	e_2	e_3
v	1	0	1
u	1	1	0
w	0	1	1

Representation- Adjacency Matrix

- There is an $N \times N$ matrix, where $|V| = N$, the Adjacent Matrix ($N \times N$) $A = [a_{ij}]$

For undirected graph

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G \\ 0 & \text{otherwise} \end{cases}$$

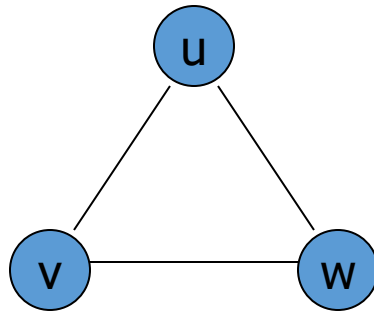
- **For directed graph**

$$a_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \text{ is an edge of } G \\ 0 & \text{otherwise} \end{cases}$$

- This makes it easier to find subgraphs, and to reverse graphs if needed.

Representation- Adjacency Matrix

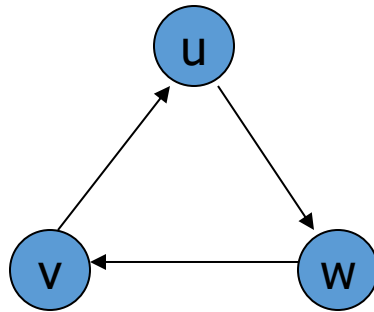
- Example: Undirected Graph $G(V, E)$



	v	u	w
v	0	1	1
u	1	0	1
w	1	1	0

Representation- Adjacency Matrix

- Example: directed Graph $G(V, E)$

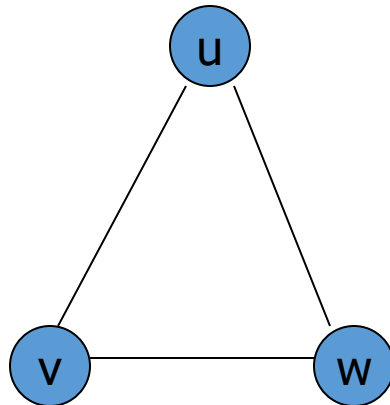


	v	u	w
v	0	1	0
u	0	0	1
w	1	0	0

Representation- Adjacency List

Each node (vertex) has a list of which nodes (vertex) it is adjacent

Example: undirected graph $G(V, E)$



node	Adjacency List
u	v , w
v	w, u
w	u , v

Euler's Totient Function

- An important quantity in number theory is ***Euler's Totient Function***:
 - The number of positive integers less than n , that are relatively prime to n .
- It is written $\phi(p)$:
 - $\phi(1) = 1$
 - $\phi(p) = p - 1$ (for p prime)
 - $\phi(m) < m - 1$ (for m composite)
- In other words, Euler's Totient Function $\phi(p)$ is ***the number of elements in $\mathbb{Z}_n^*$***

Properties of Euler's Totient Function

- Assume we have two distinct prime numbers, p and q , and an integer $n = pq$
 - Then: $\varphi(n) = \varphi(pq) = \varphi(p) \times \varphi(q) = (p-1) \times (q-1)$
 - The set of residues in $\mathbf{Z}_n$ is $\{0, 1, \dots, (pq-1)\}$
 - The residues that are not relatively prime to n are:
 - The set $\{p, 2p, \dots, (q-1)p\}$, the set $\{q, 2q, \dots, (p-1)q\}$, and 0
 - So:

$$\begin{aligned}\varphi(n) &= pq - [(q-1) + (p-1) + 1] \\ &= pq - (p + q) + 1 \\ &= (p-1) \times (q-1) \\ &= \varphi(p) \times \varphi(q)\end{aligned}$$

This fact laid the foundation to various modern encryption algorithms, including the RSA public key encryption (1977).